

fesim

User and Technical Manual

Linked employer–employee panel simulation in Stata

Johannes F. Schmieder
Boston University

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This manual explains the public command, all eight available presets, and the data and diagnostics they return. Reproducible examples illustrate wage dispersion, sorting, pay gaps, search dynamics, and wage bargaining. The technical appendix documents the implemented random processes and derives the canonical Burdett–Mortensen equilibrium, its finite-firm approximation, and the exact continuous-time worker simulation. A further appendix derives CPV values, contract wages, incumbent raises, and joint stationary histories.

This manual accompanies the unpublished 1.1.0-rc.1 candidate. Qualification is limited to Stata/MP 19 on macOS Apple Silicon; exact source and archive evidence are recorded in the repository release-review report. The stable v0.1.0 release contains the earlier simple-AKM command.

<https://github.com/johannes-schmieder/fesim>
<https://johannes-schmieder.com/>

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1 What the command produces

fesim creates synthetic linked employer–employee data. Each row is a worker observed in a retained output period, including periods of nonemployment. Worker identifiers stay fixed; employers can change; wages are observed only when employed. The command separates the economic DGP, its numerical preset, the destination network, and the observation frequency. It supports teaching, Monte Carlo experiments, and evaluation of estimators whose behavior depends on mobility and worker–firm sorting.

The installed engine uses official Stata and Mata only. Stata 19 is the supported minimum. The current development source has been qualified on Stata/MP 19 on macOS Apple Silicon; Windows qualification of the older v0.1.0 release does not qualify the later BM or CPV implementations. Reproducibility claims refer to a fixed package source, seed, command, and qualified Stata environment.

1.1 Choose a DGP

Family / preset	Short name	Purpose and interpretation
akm / simple	akmsimple	Additive worker and firm effects, independent attraction weights, and discrete interval mobility. Stylized teaching and testing baseline.
akm / stylized	akmempirical	Monthly mobility with latent worker types, firm quality, duration dependence, and sorting. All defaults are stylized; the alias names the mechanism.
akm / germany_chk_2002_2009	None	The monthly engine with selected West German wage-dispersion and sorting targets from Card, Heining, and Kline (CHK).
akmpaygap / simple	None	Two groups with different worker distributions, premium schedules, mobility probabilities, and sorting. Stylized.
akmpaygap / cck2016	None	Targets selected Card, Cardoso, and Kline (CCK) group moments and male-reference firm decomposition under the package normalization.
bm / simple	bmsimple	Homogeneous-worker, common-productivity wage posting with unemployed and on-the-job search. Exact continuum model; stylized primitives.
cpv / simple	None	Sequential wage bargaining, incumbent raises, heterogeneous firms and homogeneous workers. Stylized primitives.
cpv / heterogeneous	None	The same bargaining engine with mean-one lognormal worker ability. No endogenous ability sorting.

There are four canonical family names and eight family/preset combinations. The three aliases resolve to designated presets. Omitting `dgp()` and `preset()` selects `akm/simple`. AKM terminology follows Abowd, Kramarz, and Margolis [1]. The search model follows Burdett and Mortensen's wage-posting framework [2]; its specific discounted-worker reservation condition and implementation are derived in Appendix E.

1.2 Install and begin

To use all designs in this manual, install the development package from the repository root in Stata. A checkout also permits source-based use:

```
* Change this path to your local checkout.
cd "/path/to/fesim"
adopath ++ "."
help fesim
fesim version
fesim list
fesim presets akm
fesim describe bm, preset(simple)
```

The installed help contains numbered *click to run* links. Each invokes `fesim_run`, runs a whole example, and restores the caller's dataset. The source-only loader can compile packaged Mata sources when a compatible library is absent. The README gives development and immutable stable-tag installation instructions.

2 Syntax and options

```
fesim [, dgp(name) preset(name)
    workers(#) firms(#) periods(#)
    frequency(year|quarter|month) start(date) seed(#)
    initial(stationary|random|allunemployed) burnin(#)
    jobrule(end) truth(none|basic|full)
    connectivity(keep|largest) network(random|blocks|bridges|ladder)
    parameters(name value [name value ...]) report noreport clear]

fesim version
fesim list
fesim presets [dgp]
fesim describe dgp [, preset(name)]
```

The display lists alternatives: `report` and `noreport` are mutually exclusive, and not every initial state or network is available for every family. Use full option spellings in reproducible scripts. `fesim` generates a population; it does not accept a variable list or use the current dataset as an economic input.

2.1 Common controls

Option	Meaning
<code>dgp(name)</code> , <code>preset(name)</code>	Resolve a family/alias and a preset. Names are case-insensitive. fesim describe reports defaults, bounds, and provenance.
<code>workers(n)</code> , <code>firms(n)</code>	Generated workers and firm universe. Inactive firms need not appear. Filtering may reduce returned workers without changing requested parameters.
<code>periods(n)</code>	Retained snapshots per worker. Monthly <code>periods(60)</code> covers five years. Workers times periods cannot exceed 2,147,483,647; memory or edition limits may bind sooner.
<code>frequency(...)</code>	Output interval: year (default), quarter, or month. Changing frequency does not reinterpret annual input rates.
<code>start(date)</code>	First retained snapshot label: 2000, 2000q1, or 2000m1, for example. Default is the first period of 2000, except Germany and CCK presets, which use 2002.
<code>seed(n)</code>	Integer from 0 to 2,147,483,647. Explicit seeds preserve caller RNG state. If omitted, one caller RNG draw supplies the master seed.
<code>initial(...)</code> , <code>burnin(n)</code>	Initial state and discarded pre-sample time. Availability and units are model-specific; see Table 1.
<code>jobrule(end)</code>	Employer at the output endpoint. Dominant- and longest-employer rules are not public interfaces.
<code>truth(...)</code>	none suppresses latent columns; basic (default) adds wage components; full adds model-specific quantities. Economic draws are invariant to this choice.
<code>connectivity(...)</code>	keep (default) returns all workers. largest retains complete histories in the selected observed component. force is unavailable.
<code>network(...)</code>	Destination rule. random is the default and the only allowed design for pay-gap, BM and CPV. All four designs are available for AKM.
<code>parameters(...)</code>	Whitespace-separated scalar name/value pairs. Model and network controls belong here. Unknown, duplicate, irrelevant, conflicting, or inadmissible inputs fail validation. Numeric common controls also accept this input route; named options are preferred and duplicate specification is rejected.
<code>report</code> , <code>noreport</code>	Show (default) or suppress the report without changing data.
<code>clear</code>	Authorize replacement of loaded data. Without it, the command rejects a run that would replace the caller's dataset.

Table 1: Population, timing, and initialization defaults

Preset	Workers	Firms	Periods	Initial	Burn-in
AKM simple	10,000	500	10	stationary	0 intervals
AKM stylized	10,000	500	10	random	5 years
AKM Germany	10,000	1,000	8	random	5 years
Pay-gap simple	10,000	500	10	random	5 years
Pay-gap CCK	10,000	1,000	8	random	5 years
BM simple	10,000	500	10	stationary	0 years
CPV (both)	10,000	500	10	stationary	0 years

All presets support `initial(allunemployed)`. Exact stationary starts are available for simple AKM, BM and both CPV presets. Random starts require at least one burn-in output interval for simple AKM and at least one year for monthly AKM and pay-gap. Those AKM/pay-gap burn-in inputs are integers. BM and CPV burn-in is continuous years and can be fractional. Pay-gap uses output-clock steps within its year-valued burn-in.

Discovery and compatibility. `fesim describe` also reports observed and truth-variable inventories, applicable initialization and network choices, parameter units, and scientific source/target scope. Full-truth inventories list additions beyond basic truth; conditional network fields are separate. Existing v1 option meanings, defaults, output semantics, and named matrices are compatibility commitments. Numerical fixes that change seeded output must be disclosed. See [the compatibility policy](#) for accepted option abbreviations and future deprecation rules.

2.2 Changing model parameters

```
preserve
fesim, dgn(akm) preset(simple) workers(1000) firms(50) ///
    periods(8) seed(12345) ///
    parameters(sd_worker .45 sd_firm .20 p_ee .25) clear
matrix list r(parameters)
restore
```

The matrix has `value`, `default`, `lower`, and `upper` columns; cross-parameter restrictions also apply. Common scalar dimensions use named options. Model overrides change `stylized` to `stylized_modified`, or `targeted` to `targeted_modified`. Inspect `r(calibration_class)` and dataset characteristics when documenting an experiment.

3 DGP parameter reference

3.1 Simple AKM

Employed log wages satisfy

$$y_{it} = \mu + \alpha_i + \psi_{J_{it}} + \gamma t_{\text{years}} + \epsilon_{it}. \quad (1)$$

Worker effects, firm effects, attraction, and wage errors use separate primitives. Default random destinations do not condition on ability or pay.

Parameter	Meaning	Default
<code>mu</code>	Mean log-wage intercept	3
<code>sd_worker</code>	Worker-effect SD	.40
<code>sd_firm</code>	Firm-effect SD	.15
<code>sd_error</code>	Idiosyncratic log-wage SD	.20
<code>firm_size_sd</code>	Log attraction scale	1
<code>p_eu, p_ee, p_ue</code>	Annual probabilities	.08, .12, .60
<code>wage_trend</code>	Log points per retained-sample year	0

SDs are nonnegative. EU and EE probabilities are nonnegative and sum to less than one; UE lies in $[0, 1]$. Positive EE requires at least two firms. Stationary initialization is undefined when both EU and UE are zero.

3.2 Monthly AKM: stylized and Germany

The stylized preset retains the wage and attraction defaults above but replaces interval probabilities by monthly transitions from annual hazards.

Parameter	Stylized default	Role
<code>rho_z_alpha, rho_q_psi</code>	.30, .50	Latent correlations
<code>kappa_eu</code>	−2.416230719	EU log-hazard intercept
<code>eu_worker, eu_firm, eu_duration</code>	.10, −.10, −.20	EU type, quality, duration slopes
<code>kappa_ee</code>	−2.010765611	EE log-hazard intercept
<code>ee_worker, ee_firm, ee_duration</code>	.10, −.10, −.15	EE type, quality, duration slopes
<code>kappa_ue</code>	−.087421572	UE log-hazard intercept
<code>ue_worker, ue_duration</code>	.15, −.25	UE slopes
<code>theta_sort, theta_quality</code>	.25, .10	Sorting / quality preference
<code>theta_up, theta_down</code>	.20, −.10	Asymmetric EE distances

Exact intercepts are the logs of $-\log(1 - .08 - .12)$ split in .08/.12 proportions and of $-\log(1 - .60)$ for UE. Monthly presets do not accept `p_eu`, `p_ee`, or `p_ue`. Correlations lie in $[-1, 1]$; a zero wage-effect SD requires its corresponding latent correlation to be zero. Other numerical bounds are available through `fesim describe`.

The Germany preset changes `sd_worker` to .357, `sd_firm` to .230, `sd_error` to .135, and `theta_sort` to 2.2. The first three values are CHK component targets; sorting targets employed-observation covariance .0205, half the reported 2 Cov = .041 [3]. Hazards and durations remain stylized. This is a targeted wage-dispersion and sorting design, not a reconstruction of German employment histories.

3.3 Pay-gap presets

Men are `group=0`, women `group=1`; gaps are men minus women. Each firm has surplus $s_j \sim N(0, 1)$ and group premium

$$\psi_{gj} = a_g + b_g s_j + \tau_g \eta_{gj}, \quad y_{it} = \mu_g + \alpha_i + \psi_{gJ_{it}} + \gamma_g t_{\text{years}} + \epsilon_{it}.$$

Surplus is not restandardized in the finite sample. Both schedules exist at each firm, even when only one group is observed there.

Parameter (men, women where paired)	Simple	CCK
<code>female_share</code>	.50	.46
<code>mu_m, mu_f</code>	3, 2.9	3, 2.815
<code>sd_worker_m, sd_worker_f</code>	.40, .40	.420, .400

Parameter (men, women where paired)	Simple	CCK
premium_intercept_m,	.05, .03	.113, .099
premium_intercept_f		
premium_loading_m, premium_loading_f	.15, .12	.247, .12567
premium_deviation_sd_m,	0, .09	0, .1719768912
premium_deviation_sd_f		
sd_error_m, sd_error_f	.20, .20	.143, .125
firm_size_sd	1	1
p_eu_m, p_eu_f	.08, .08	.08, .08
p_ee_m, p_ee_f	.12, .12	.12, .12
p_ue_m, p_ue_f	.60, .60	.60, .60
group_sort_m, group_sort_f	.30, 0	.142, 0
worker_sort_m, worker_sort_f	.30, .30	.18, .273
wage_trend_m, wage_trend_f	0, 0	.05, .05

Groupwise competing-probability restrictions are the same as in simple AKM; at least two firms are required. `female_share` lies in $[0, 1]$. Group moments are undefined when the relevant group/sample is empty. The CCK preset targets selected group wage/effect moments and a male-reference decomposition [4]. Its standard-normal-surplus normalization differs from CCK's low-surplus-firm normalization. Targets are not constraints forcing each realized sample to reproduce published moments.

3.4 Burdett–Mortensen

Parameter	Economic meaning	Default	Unit
<code>b</code>	Unemployment flow payoff	.4	Wage level
<code>p</code>	Common productivity	1	Wage level
<code>lambda_u</code>	Unemployed offer rate	1	Annual rate
<code>lambda_e</code>	Employed offer rate	.5	Annual rate
<code>delta</code>	Job destruction rate	.2	Annual rate
<code>r</code>	Worker discount rate	.05	Annual rate
<code>random_firms</code>	0 midpoint / 1 random quantiles	0	Indicator

Rates must be positive and at most 1,000. Productivity must exceed b , and solved wage support must be positive because output wages are logs. Support and residual checks can reject numerically unresolved inputs. There is no public control for measurement error, heterogeneous productivity, bargaining, or firm attraction.

The continuum equilibrium is solved first. Finite posted wages discretize its offer distribution. Workers contact finite firms uniformly and accept only strictly higher wages while employed. The exact finite stationary allocation differs from the continuum worker distribution; a finite wage-posting game is not solved anew (Appendix E).

3.5 Cahuc–Postel-Vinay–Robin bargaining

`dgp(cpv)` implements the sequential bargaining protocol of Cahuc, Postel-Vinay, and Robin [6] in a finite firm economy. `preset(simple)` has identical workers; `preset(heterogeneous)` adds

mean-one lognormal worker ability. Both have heterogeneous firms. These are stylized designs, not estimates or a replication of the paper.

Parameter	Meaning	Default	Unit
b	Unemployment flow payoff	1	Per efficiency unit
p_min, p_max	Uniform firm productivity support	1.5, 2	Per efficiency unit
lambda_u	Unemployed offer rate	.5	Per year
lambda_e	Employed offer rate	.3	Per year
delta	Destruction rate	.2	Per year
r	Discount rate	.05	Per year
beta	Worker bargaining share	.5	Share in $[0, 1]$
sd_worker	Log ability standard deviation	0 or .4	Log units
random_firms	Midpoint (0) or random (1) quantiles	0	Indicator

The heterogeneous preset changes only **sd_worker** to .4. All model controls belong in **parameters()**, for example:

```
fesim, dgp(cpv) preset(heterogeneous) workers(5000) firms(100) ///
    periods(60) frequency(month) seed(12345) truth(full) ///
    parameters(beta .3 sd_worker .5 lambda_e .4) clear
```

Firm productivity must be positive with $p_{\max} \geq p_{\min}$; equal endpoints are allowed. The unemployed offer, destruction, and discount rates are positive and at most 1,000; the employed offer rate may be zero. The flow payoff is positive and ability dispersion is nonnegative. Every generated firm must have strictly positive full surplus and entry wage. The solver rejects an invalid economy without clipping wages or removing firms. Random firm draws are included in transactional rollback.

Defaults are 10,000 workers, 500 firms, ten annual observations, **initial(stationary)** and zero burn-in for both presets. Stationary initialization draws the joint employer, contract and tenure state. **initial(random)** starts half the workers employed at random firms with entry contracts and requires positive continuous-year burn-in. **initial(allunemployed)** starts all workers with zero unemployment age. Only **network(random)** and **jobrule(end)** are supported; **connectivity(keep)** and **connectivity(largest)** are available.

Outside offers can raise the wage without changing the employer or spell. A move to a more productive firm can reduce today's wage because future bargaining opportunities improve. Worker ability multiplies production, wages, unemployment income and values; common contact rates imply no ability sorting. Neither structural firm productivity nor a contract wage is an AKM firm effect. Appendix F derives the implementation.

4 Network designs and connectedness

network(random) means the selected DGP's baseline destination mechanism; empirical-AKM and pay-gap sorting can still be nonzero. In BM and CPV it means uniform contacts with structural acceptance.

blocks. Independently assign workers and firms to balanced communities. Entry/initialization refer to the worker's home block; EE moves refer to the current firm's block. Same-block destination mass is multiplied by $\exp(\text{block_log_bonus})$. Defaults: **block_count**=4 and **block_log_bonus**= $\ln(9)$. Zero bonus exactly nests random destinations.

bridges. Ordinary moves stay strictly within blocks. Redirect an exact number of eligible retained EE destinations across adjacent blocks: $1 \rightarrow 2, 2 \rightarrow 3, \dots, B-1 \rightarrow B$, then repeat. Distinct workers are used. `bridge_count` defaults to $B-1$. Insufficient eligible events cause failure; at least two firms per block are required. `block_log_bonus` is inapplicable.

ladder. Rank firms by persistent wage effects using tied percentile midranks. Candidates within `ladder_band=.10` of the origin are lateral; lower/higher candidates outside the band are downward/upward. Defaults `ladder_down_share=.10`, `ladder_lateral_share=.20`, and `ladder_up_share=.70` are renormalized over feasible directions. Shares sum to one; ordinary destination weights apply within direction.

These stress designs apply only to AKM. The ladder is a reduced-form experiment, not a structural search equilibrium. Network scalars belong inside `parameters()`.

The observed bipartite graph has ever-employed workers and active firms as nodes, and one edge per unique observed match. `connectivity(largest)` selects by employed observations, then workers, firms, and smallest component ID. Complete selected worker histories, including unemployment, are retained. Never-employed workers are excluded; IDs are not renumbered.

`r(network)` has 21 rows and `generated/returned` columns. It includes an undirected firm-mobility graph based on adjacent observed direct moves. Firm links, edge-weight percentiles, articulation firms, and graph bridges are descriptive diagnostics.

`r(leaveout)` separately audits the largest observed bipartite component. Worker deletion removes complete histories. The audit repeatedly removes articulation workers and selects the largest survivor until a robust set or emptiness. This extends the worker-set logic in leave-out estimation [5]. A match is vulnerable only if deletion separates firms; isolating a stayer alone does not count. This audit does not further filter public data.

5 Reading the panel and results

5.1 Observed variables

Variable	Interpretation
<code>workerid, time</code>	Unique worker-period key, sorted and balanced within returned workers. Time is Stata annual, quarterly, or monthly.
<code>firmid, employed</code>	Current firm and 0/1 employment. Firm ID is missing in nonemployment.
<code>lnwage</code>	Log wage, missing when unemployed. BM posted wages and CPV contract wages are levels before the logarithm.
<code>spellid, tenure</code>	Employment spell and elapsed tenure in output-period units; missing when unemployed.
<code>newjob</code>	Changed spell from preceding snapshot; first-row missing. Same-firm re-entry can start a new spell.
<code>from_unemp</code>	Employment after preceding-snapshot nonemployment; first-row missing.
<code>to_unemp</code>	Employment followed by next-snapshot nonemployment; last-row missing. This indicator is forward-looking.

Variable	Interpretation
<code>jobtojob</code>	Employment at different firms in adjacent snapshots; first-row missing. It need not identify one latent EE event.
<code>ntransitions</code>	Latent state transitions in the interval; first-row missing under the common reporting convention.
<code>unemp_duration</code>	Added by monthly AKM, BM and CPV: elapsed unemployment duration in output units; missing when employed.
<code>group</code>	Pay-gap group: 0 Men, 1 Women.
<code>nbridges_imposed</code>	Imposed bridge events in an interval; present for bridges under every truth mode.

Divide durations by 1, 4, or 12 to convert output units to years. `r(durations)` already uses years. BM full truth retains exact first-interval counts even though common backward flags are first-row missing.

5.2 Truth variables

AKM and pay-gap basic truth includes `alpha_true`, `psi_true`, `time_true`, `xb_true`, `match_true`, `epsilon_true`, and `lnwage_true`. Covariate and match components are zero in current models. Pay-gap `psi_true` is the observed group's premium.

Full monthly-AKM truth adds `worker_type_true` and `firm_quality_true`; blocks/bridges add `worker_block_true` and `firm_block_true`. Full pay-gap adds `firm_surplus_true`, `psi_male_true`, and `psi_female_true`. Job-specific truth is missing when unemployed.

BM basic truth is `lnwage_true`, `posted_wage_true`, and `productivity_true`; it does not fabricate AKM worker effects. Appendix E.13 lists full structural truth.

CPV basic truth reports log and level contract wages, worker ability, firm productivity per efficiency unit, and match productivity. Full truth adds bargaining references and worker values, exact offers/raises/transitions, and employment/unemployment exposure. Worker ability remains defined during unemployment; job attributes are missing. Appendix F specifies these objects. There is no CPV structural `psi_true`.

5.3 Returned results

Result	Use
<code>r(N)</code> , <code>r(N_workers)</code> , <code>r(N_firms)</code> , <code>r(N_firms_active)</code>	Returned rows/workers, requested firms, and active firms.
<code>r(employment_rate)</code> , <code>r(p_eu_realized)</code> , <code>r(p_ue_realized)</code> , <code>r(p_ee_realized)</code>	Observed employment and endpoint probabilities using appropriate origin-state denominators.
<code>r(parameters)</code>	Resolved scalar values, defaults, bounds.
<code>r(moments)</code> , <code>r(targets)</code>	Realized moments, applicable targets, realized-minus-target differences, and relative differences.
<code>r(network)</code> , <code>r(leaveout)</code>	Graph and leave-out diagnostics.

Result	Use
<code>r(durations)</code>	Year-valued distributions for monthly AKM, BM and CPV: count, mean, sample SD, p10, p50, p90.
<code>r(bridges)</code>	Eight-column imposed-event ledger.
<code>r(group_moments)</code> , <code>r(group_targets)</code>	Pay-gap 17-by-2 matrices.
<code>r(decomposition)</code> , <code>r(decomposition_targets)</code>	Nine-by-three male, female, symmetric reference decompositions.
<code>r(solver)</code> , <code>r(bm_flows)</code> , <code>r(bm_firms)</code>	BM solution/residuals, theory/event/endpoint flows, firm summaries.
<code>r(cpv_flows)</code> , <code>r(cpv_firms)</code>	CPV finite theory, event and endpoint flows, and firm summaries; <code>r(solver)</code> reports CPV Bellman values and residuals.
<code>r(runtime_total)</code> , <code>r(runtime_solve)</code> , <code>r(runtime_simulate)</code> , <code>r(runtime_output)</code>	Total and stage timings in seconds.
<code>r(dgp)</code> , <code>r(preset)</code> , <code>r(dgp_alias)</code> , <code>r(seed)</code> , <code>r(version)</code> , <code>r(command)</code>	Identity, alias, actual master seed, version, serialized scientific configuration.

`r()` is transient. Copy a matrix before running another `r`-class command. For example, execute `matrix saved_moments = r(moments)` before `summarize lnwage`.

6 Reproducible examples and figures

Each numbered figure example is read directly from its runnable `docs/manual_examples/` do-file, so printed and executed code share one source. Examples require `fesim` on the adopath and create their own data. `preserve/restore` protects caller data; graphs remain available. These illustrative samples are not calibration or estimator tests.

6.1 Simple AKM: wages and realized sorting

```
version 19.0
preserve
fesim, dgp(akmsimple) workers(3000) firms(100) periods(8) ///
    seed(20260901) truth(basic) noreport clear
matrix list r(moments)
isid workerid time
histogram lnwage if employed, density color(navy%65) ///
    xtitle("Log wage") title("A. Employed wage distribution") ///
    name(fesim_akm_wages, replace)
keep if employed
collapse (mean) alpha_true psi_true (count) size=workerid, by(firmid)
tway scatter alpha_true psi_true [aw=size], ///
    mcolor(navy%60) msymbol(Oh) ///
    xtitle("Firm wage effect") ytitle("Mean worker effect at firm") ///
    title("B. Realized worker sorting") name(fesim_akm_sort, replace)
graph combine fesim_akm_wages fesim_akm_sort, ///
    cols(2) xsize(10) ysize(4) name(fesim_akm, replace)
restore
```

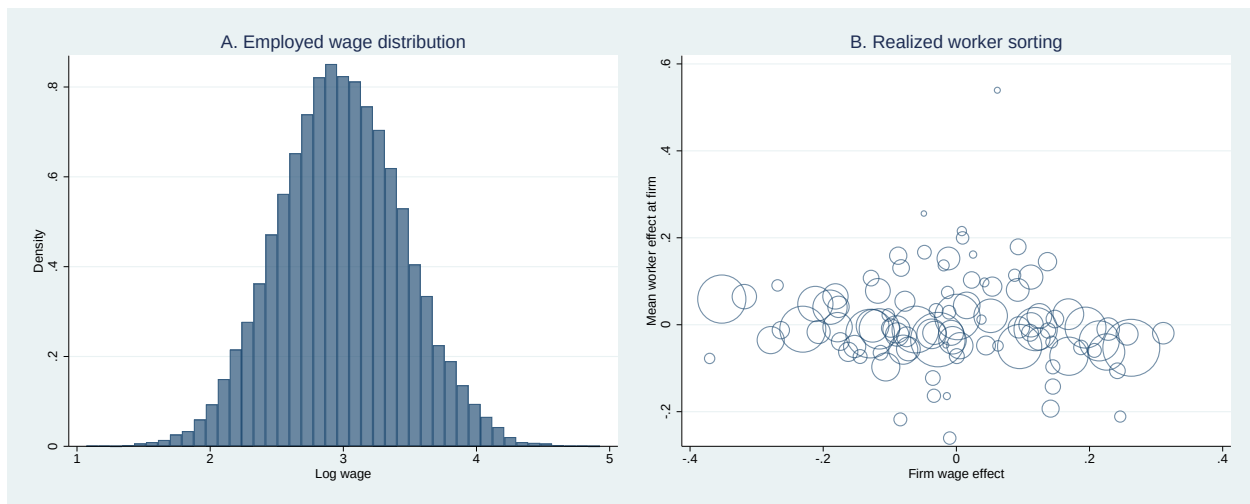


Figure 1: Simple AKM: 3,000 workers, 100 firms, eight years, seed 20260901. Left: employed log wages. Right: mean worker effect versus firm effect; marker area reflects employed-observation count. Attraction creates size heterogeneity. The random design does not impose positive population sorting, but finite-sample means vary.

6.2 Monthly mobility and Germany-targeted sorting

```

version 19.0
preserve
fesim, dgp(akmempirical) workers(3000) firms(100) periods(8) ///
    seed(20260902) truth(full) noreport clear
matrix list r(durations)
histogram ntransitions if !missing(ntransitions), discrete percent ///
    color(navy%65) xtitle("Latent monthly transitions per annual interval") ///
    title("A. Stylized mobility") name(fesim_emp_moves, replace)
fesim, dgp(akm) preset(germany_chk_2002_2009) ///
    workers(5000) firms(500) periods(8) seed(20260903) ///
    truth(full) noreport clear
matrix list r(targets)
keep if employed
collapse (mean) alpha_true psi_true (count) size=workerid, by(firmid)
tway scatter alpha_true psi_true [aw=size], ///
    msymbol(Oh) mcolor(maroon%55) ///
    xtitle("Firm wage effect") ytitle("Mean worker effect at firm") ///
    title("B. Germany-targeted sorting") name(fesim_emp_sort, replace)
graph combine fesim_emp_moves fesim_emp_sort, ///
    cols(2) xsize(10) ysize(4) name(fesim_mobility, replace)
restore

```

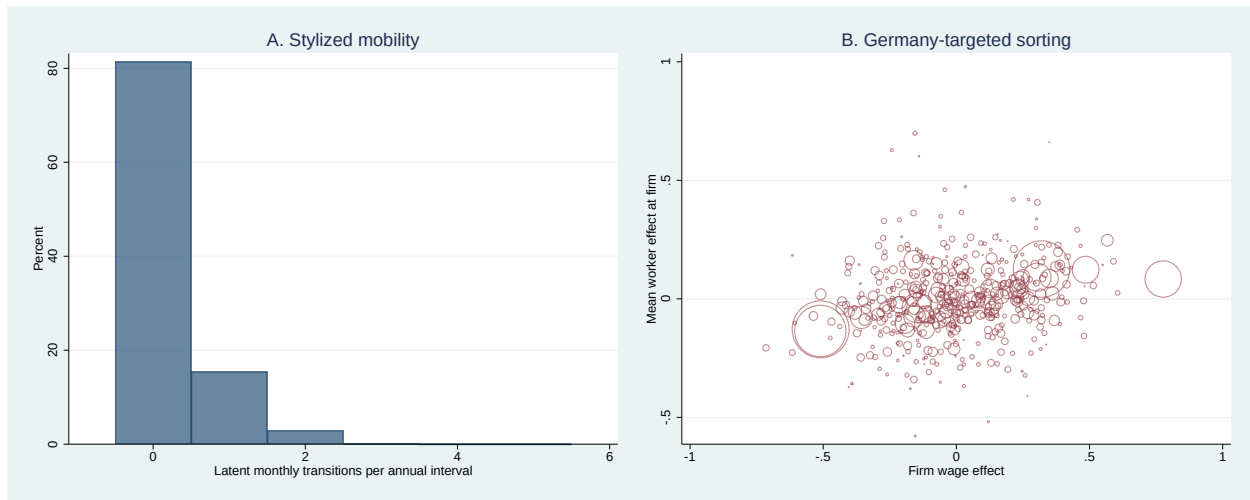


Figure 2: Two simulations. Left: latent monthly transitions per annual interval in the stylized engine (3,000 workers, 100 firms, eight years, seed 20260902), excluding first observations. Right: Germany-targeted sorting (5,000 workers, 500 firms, eight years, seed 20260903). Marker area reflects employed observations. Default five-year burn-in applies to both. The right panel illustrates sorting rather than testing calibration accuracy in this smaller sample.

6.3 Pay gaps and counterfactual firm schedules

```

version 19.0
preserve
fesim, dgp(akmpaygap) preset(cck2016) workers(5000) firms(200) ///
    periods(8) seed(20260904) truth(full) noreport clear
matrix list r(decomposition)
twoway (kdensity lnwage if employed & group==0, lcolor(navy)) ///
    (kdensity lnwage if employed & group==1, lcolor(maroon)), ///
    legend(order(1 "Men" 2 "Women")) ///
    xtitle("Log wage") ytitle("Density") ///
    title("A. Employed wage distributions") name(fesim_gap_wage, replace)
keep if employed
bysort firmid: keep if _n==1
twoway (scatter psi_male_true firm_surplus_true, ///
    mcolor(navy%65) msymbol(Oh)) ///
    (scatter psi_female_true firm_surplus_true, ///
    mcolor(maroon%55) msymbol(Th)), ///
    legend(order(1 "Male schedule" 2 "Female schedule")) ///
    xtitle("Firm surplus index") ytitle("Firm premium (log points)") ///
    title("B. Both schedules at each active firm") ///
    name(fesim_gap_schedule, replace)
graph combine fesim_gap_wage fesim_gap_schedule, ///
    cols(2) xsize(10) ysize(4) name(fesim_paygap, replace)
restore

```

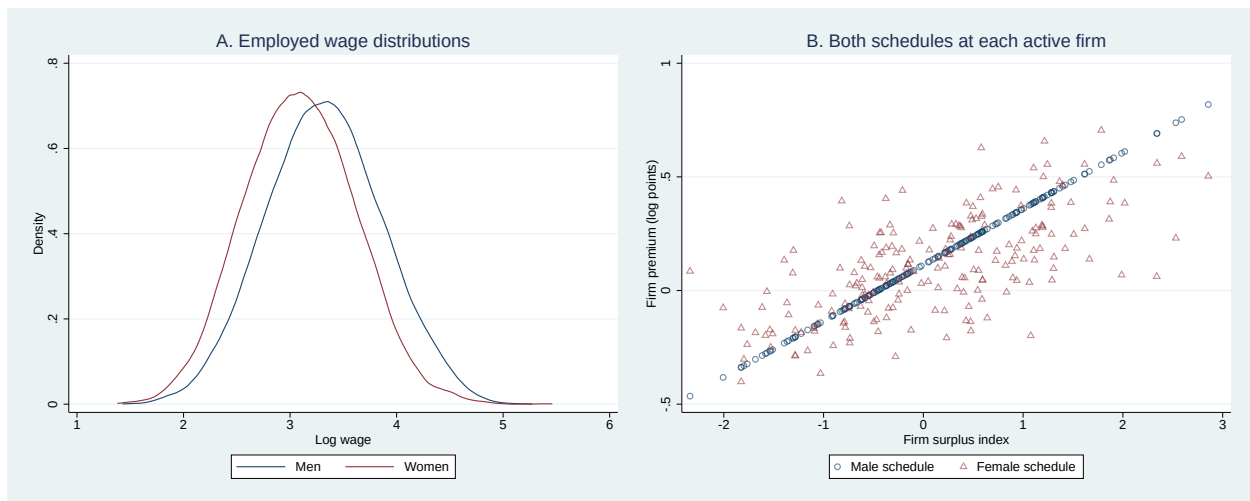


Figure 3: CCK-inspired data: 5,000 workers, 200 firms, eight annual observations, five-year burn-in, seed 20260904. Left: smoothed employed log-wage distributions. Right: one observation per active firm showing both premium schedules against the same surplus index. The female schedule has a firm-specific deviation as well as a common-surplus loading. Premiums use the package normalization.

6.4 BM wage distributions and firm size

```

version 19.0
preserve
fesim, dgp(bm) workers(10000) firms(100) periods(5) ///
    seed(20260905) truth(full) noreport clear
matrix list r(solver)
matrix list r(bm_flows)
matrix list r(bm_firms)
keep if employed & time==2004
collapse (count) size=workerid (first) posted_wage_true ///
    expected_firm_mass_true offer_quantile_true, by(firmid)
assert _N==100
sort posted_wage_true
generate double worker_cdf = sum(size)
replace worker_cdf = worker_cdf / worker_cdf[_N]
generate double offer_cdf = _n / _N
generate double G = .2*offer_quantile_true ///
    (.2+.5*(1-offer_quantile_true))
tway (line offer_cdf posted_wage_true, lcolor(gs8) lpattern(dash)) ///
    (line worker_cdf posted_wage_true, lcolor(navy)) ///
    (line G posted_wage_true, lcolor(maroon) lpattern(shortdash)), ///
    legend(order(1 "Finite firm offers" 2 "Simulated workers" ///
    3 "Continuum worker CDF") cols(1) size(small)) ///
    xtitle("Posted wage (level)") ytitle("Cumulative share") ///
    title("A. Firm and worker wage distributions") ///
    name(fesim_bm_cdf, replace)
generate double expected_size = 10000*expected_firm_mass_true
tway (scatter size posted_wage_true, mcolor(navy%65) msymbol(Oh)) ///
    (line expected_size posted_wage_true, lcolor(maroon)), ///
    legend(order(1 "Simulated" 2 "Finite stationary expectation")) ///
    xtitle("Posted wage (level)") ytitle("Workers at final snapshot") ///
    title("B. Higher wages attract larger workforces") ///
    name(fesim_bm_size, replace)
graph combine fesim_bm_cdf fesim_bm_size, ///
    cols(2) xsize(10) ysize(4.5) name(fesim_bm, replace)
restore

```

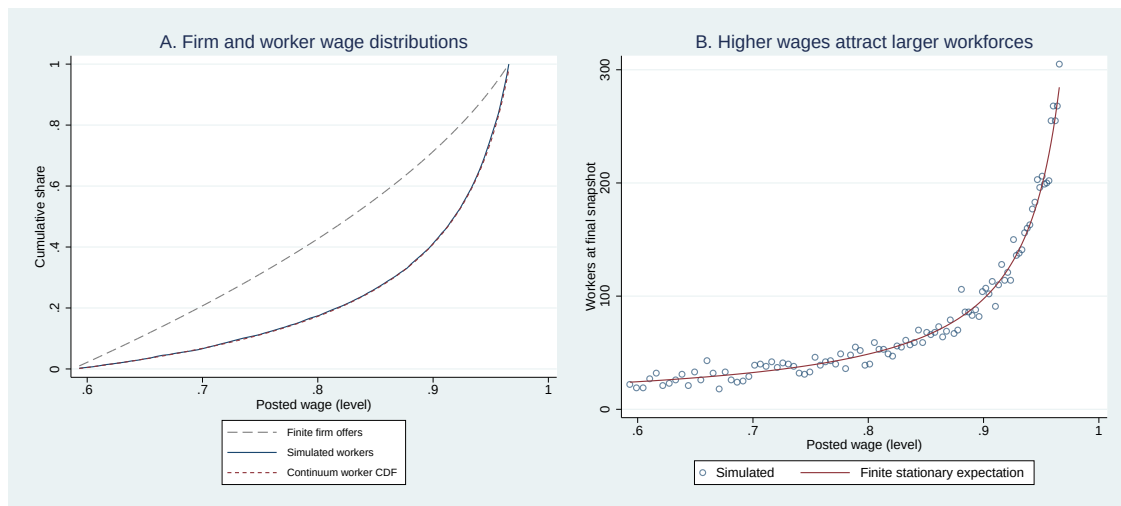


Figure 4: Default BM primitives: 10,000 workers, 100 midpoint firms, five annual observations, stationary start, seed 20260905. Both panels use the last snapshot. All 100 firms are active in this draw (checked in code). Lines connect CDF values at finite wage points. Workers concentrate at higher wages than firm offers, and sizes fluctuate around exact finite stationary expectations. The continuum worker CDF is a separate benchmark.

6.5 BM worker histories

```

version 19.0
preserve
fesim, dgp(bm) workers(300) firms(100) periods(60) ///
    frequency(month) start(2000m1) seed(20260906) ///
    truth(full) noreport clear
keep if workerid<=4
generate double year = (time-tm(2000m1)+1)/12
generate double plotted_wage = posted_wage_true
replace plotted_wage = 0 if !employed
twoway connected plotted_wage year, sort msymbol(o) msize(vsmall) ///
    lcolor(navy) mcolor(navy) ///
    by(workerid, cols(2) note("Zero denotes unemployment, not a wage.)) ///
    xtitle("Years since retained-sample boundary") ///
    ytitle("Posted wage (level)") ylabel(0 .4 .6 .8 1) ///
    xsize(10) ysize(6) name(fesim_bm_paths, replace)
restore

```



Figure 5: First four worker IDs: 300 workers, 100 firms, 60 monthly snapshots, stationary start, seed 20260906. Zero denotes unemployment only in the plot; nonemployment wages remain missing in the data. Connected points are monthly observations, not exact event times. A worker can separate and re-enter at a lower wage between snapshots even though every accepted latent EE move is upward.

6.6 BM events versus endpoint probabilities

```

version 19.0
preserve
fesim, dgp(bm) workers(4000) firms(100) periods(5) ///
    frequency(year) seed(20260907) truth(none) noreport clear
matrix annual = r(bm_flows)
fesim, dgp(bm) workers(4000) firms(100) periods(60) ///
    frequency(month) seed(20260907) truth(none) noreport clear
matrix monthly = r(bm_flows)
assert reldif(annual[2,2],monthly[2,2])<1e-10
assert reldif(annual[3,2],monthly[3,2])<1e-10
assert reldif(annual[4,2],monthly[4,2])<1e-10
matrix comparisons = (annual[2..4,1], annual[2..4,2], ///
    annual[2..4,4], monthly[2..4,4])
matrix colnames comparisons = theory event annual_endpoint monthly_endpoint
clear
svmat double comparisons, names(col)
generate byte flow = _n
label define fesim_flow 1 "UE" 2 "EU" 3 "EE", replace
label values flow fesim_flow
graph bar theory event annual_endpoint monthly_endpoint, over(flow) ///
    bar(1,color(gs10)) bar(2,color(navy)) ///
    bar(3,color(maroon)) bar(4,color(teal)) ///
    legend(order(1 "Finite stationary hazard" 2 "Exact event / exposure" ///
    3 "Annual endpoint probability" 4 "Monthly endpoint probability x 12") ///
    cols(2) size(small)) ytitle("Rate or probability per year") ///
    xsize(10) ysize(5) name(fesim_bm_flows, replace)
matrix drop annual monthly comparisons
restore

```

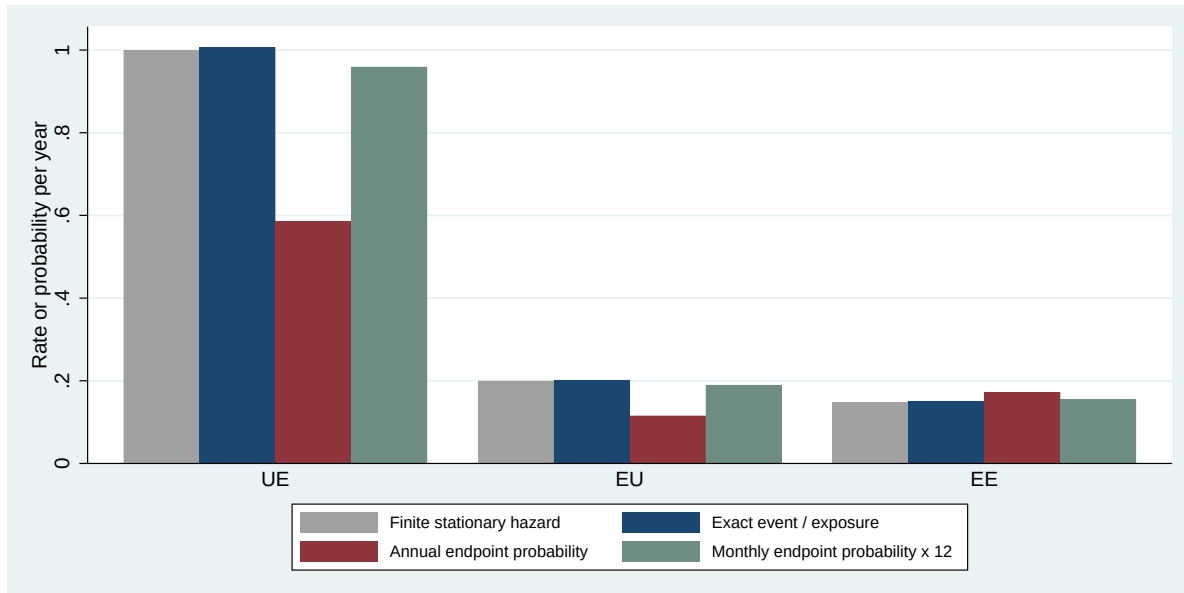


Figure 6: BM: 4,000 workers, 100 firms, five years, stationary start, seed 20260907. Annual and monthly runs use the same continuous history. Event rates divide accepted transitions by years at risk and agree across frequencies (asserted). Endpoint probabilities measure different objects. Multiplying monthly probabilities by 12 is descriptive annualization, not recovery of a continuous hazard. Observed firm changes can include intervening unemployment.

6.7 CPV histories: incumbent raises and wage cuts

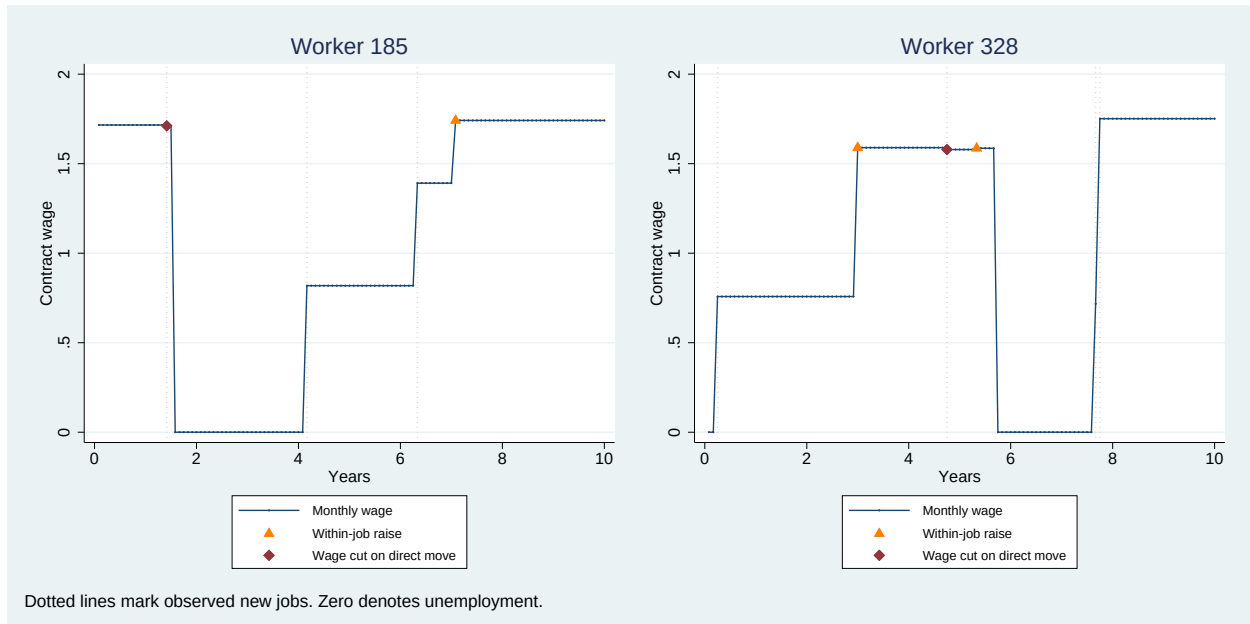


Figure 7: Two workers selected to illustrate direct-move wage cuts from 1,000 workers, 100 firms and 120 monthly snapshots; seed 20260908, $\beta = .15$, otherwise simple-preset defaults. Dotted lines identify observed new jobs, triangles mark within-job raises, and diamonds mark wage cuts in intervals containing exactly one direct EE and no EU or UE. These are selected illustrations, not estimates of event frequencies.

```

version 19.0
preserve
fesim, dgp(cpv) workers(1000) firms(100) periods(120) ///
    frequency(month) seed(20260908) truth(full) ///
    parameters(beta .15) noreport clear
generate double year=(time-tm(2000m1)+1)/12
by workerid: generate byte cut=jobtojob==1 & n_ee_true==1 & ///
    n_eu_true==0 & n_ue_true==0 & lnwage<lnwage[_n-1]
by workerid: generate byte raise=_n>1 & employed & ///
    spellid==spellid[_n-1] & lnwage>lnwage[_n-1]+1e-12
by workerid: egen byte has_cut=max(cut)
by workerid: egen byte has_raise=max(raise)
quietly levelsof workerid if has_cut & has_raise, local(candidates)
local first : word 1 of `candidates'
local second : word 2 of `candidates'
assert "`second'"!=""
generate double wage=cond(employed,contract_wage_true,0)
local panel=0
foreach id in `first' `second' {
    local ++panel
    quietly levelsof year if workerid==`id' & newjob==1, local(boundaries)
    twoway (connected wage year if workerid==`id', ///
        msize(vtiny) lcolor(navy) mcolor(navy)) ///
        (scatter wage year if workerid==`id' & raise, ///
            msymbol(T) mcolor(orange)) ///
        (scatter wage year if workerid==`id' & cut, ///
            msymbol(D) mcolor(maroon)), ///
        xline(`boundaries', lcolor(gs12) lpattern(dot)) ///

```

```

    title("Worker `id'") xtitle("Years") ytitle("Contract wage") ///
    legend(order(1 "Monthly wage" 2 "Within-job raise" ///
    3 "Wage cut on direct move") cols(1) size(small)) ///
    name(fesim_cpv_path`panel', replace)
}
graph combine fesim_cpv_path1 fesim_cpv_path2, cols(2) ///
    note("Dotted lines mark observed new jobs. Zero denotes unemployment.") ///
    xsize(10) ysize(5) name(fesim_cpv_paths, replace)
restore

```

6.8 CPV wage dispersion within firms

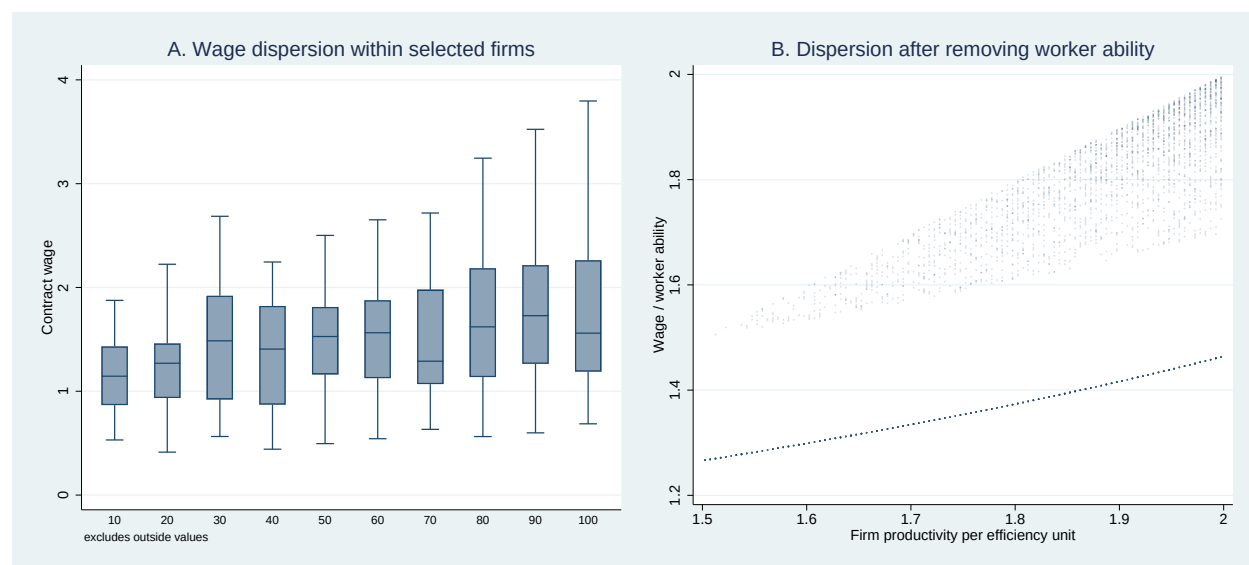


Figure 8: Final annual snapshot of 6,000 workers and 100 firms over three years, heterogeneous preset, seed 20260909. The left panel shows selected firms' wage distributions (outside values omitted from the display). The right panel divides each wage by worker ability: bargaining history still produces dispersion among workers at the same firm.

```

version 19.0
preserve
fesim, dgp(cpv) preset(heterogeneous) workers(6000) firms(100) ///
    periods(3) seed(20260909) truth(full) noreport clear
keep if employed & time==2002
graph box contract_wage_true if mod(firmid,10)==0, ///
    over(firmid, label(labsize(small))) nooutsides ///
    title("A. Wage dispersion within selected firms") ///
    ytitle("Contract wage") name(fesim_cpv_box, replace)
generate double efficiency_wage=contract_wage_true/worker_ability_true
twoway scatter efficiency_wage firm_productivity_true, ///
    msymbol(Oh) msize(vtiny) mcolor(navy%25) ///
    title("B. Dispersion after removing worker ability") ///
    xtitle("Firm productivity per efficiency unit") ///
    ytitle("Wage / worker ability") name(fesim_cpv_normalized, replace)
graph combine fesim_cpv_box fesim_cpv_normalized, cols(2) ///
    xsize(10) ysize(4.5) name(fesim_cpv_dispersion, replace)
restore

```

6.9 Changing bargaining power

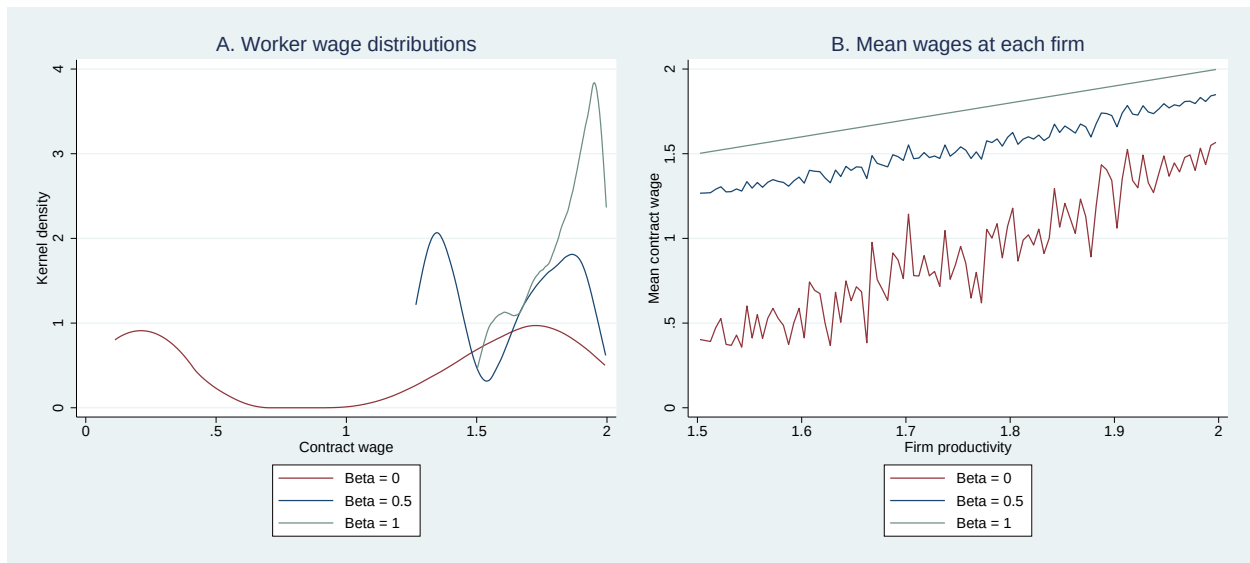


Figure 9: Final snapshot of 4,000 homogeneous workers and 100 firms over five years, seed 20260910, at three bargaining shares. Firm and employment paths are held fixed by the seed. Wages change with bargaining power; at $\beta = 1$ workers receive match output and have no incumbent raises. Kernel curves smooth a discrete finite-firm distribution.

```
version 19.0
preserve
tempfile comparisons
foreach beta in 0 .5 1 {
    fesim, dgp(cpv) workers(4000) firms(100) periods(5) ///
        seed(20260910) truth(full) parameters(beta `beta') noreport clear
    matrix list r(cpv_flows)
    keep if employed & time==2004
    generate double bargaining=`beta'
    keep bargaining firmid firm_productivity_true contract_wage_true
    if `beta'!=0 append using `comparisons'
    save `comparisons', replace
}
twoway (kdensity contract_wage_true if bargaining==0, lcolor(maroon)) ///
    (kdensity contract_wage_true if bargaining==.5, lcolor(navy)) ///
    (kdensity contract_wage_true if bargaining==1, lcolor(teal)), ///
    legend(order(1 "Beta = 0" 2 "Beta = 0.5" 3 "Beta = 1") cols(1)) ///
    title("A. Worker wage distributions") xtitle("Contract wage") ///
    ytitle("Kernel density") name(fesim_cpv_density, replace)
collapse (mean) contract_wage_true (first) firm_productivity_true, ///
    by(bargaining firmid)
twoway (line contract_wage_true firm_productivity_true if bargaining==0, ///
    sort lcolor(maroon)) (line contract_wage_true firm_productivity_true ///
    if bargaining==.5, sort lcolor(navy)) ///
    (line contract_wage_true firm_productivity_true if bargaining==1, ///
    sort lcolor(teal)), legend(order(1 "Beta = 0" 2 "Beta = 0.5" ///
```

```

3 "Beta = 1") cols(1)) title("B. Mean wages at each firm") ///
xtitle("Firm productivity") ytitle("Mean contract wage") ///
name(fesim_cpv_mean, replace)
graph combine fesim_cpv_density fesim_cpv_mean, cols(2) ///
xsize(10) ysize(4.5) name(fesim_cpv_bargaining, replace)
restore

```

6.10 An AKM projection and a mover event study

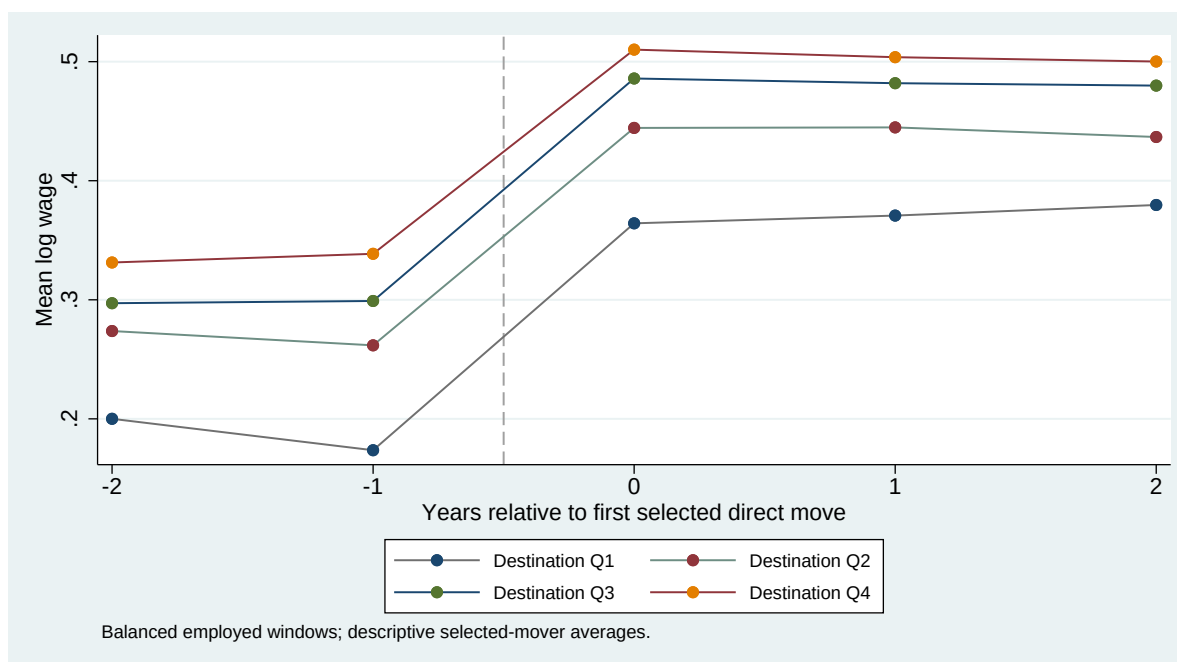


Figure 10: Heterogeneous CPV preset: 5,000 workers, 100 firms, ten annual observations, seed 20260911. Workers are aligned around their first selected direct move and must be employed at all five snapshots. Destination groups are productivity quartiles. These selected-mover averages are descriptive; they are not causal employer effects.

The example uses official `areg` to absorb worker effects while including firm and time indicators. This is an additive projection of history-dependent wages. Its firm coefficients should not be interpreted as the primitive productivity levels. There is no population AKM projection API in this version.

```

version 19.0
preserve
fesim, dgp(cpv) preset(heterogeneous) workers(5000) firms(100) ///
    periods(10) seed(20260911) truth(full) noreport clear
* Official Stata AKM-style projection; not structural productivity recovery.
areg lnwage i.firmid i.time if employed, absorb(workerid)
* Select the first observed interval with exactly one direct EE and no EU/UE.
generate double move_time=time if jobtojob==1 & n_ee_true==1 & ///
    n_eu_true==0 & n_ue_true==0
by workerid: egen double first_move=min(move_time)
generate double event_time=time-first_move
by workerid: egen byte destination=max(cond(event_time==0, ///
    ceil(4*firmid/100),.))

```

```

keep if inrange(event_time,-2,2) & employed
by workerid: keep if _N==5
assert _N>0
collapse (mean) lnwage (count) workers=workerid, by(event_time destination)
list event_time destination workers, noobs sepby(destination)
twayway (connected lnwage event_time if destination==1, lcolor(gs7)) ///
    (connected lnwage event_time if destination==2, lcolor(teal)) ///
    (connected lnwage event_time if destination==3, lcolor(navy)) ///
    (connected lnwage event_time if destination==4, lcolor(maroon)), ///
xline(-.5, lpattern(dash) lcolor(gs10)) xlabel(-2(1)2) ///
legend(order(1 "Destination Q1" 2 "Destination Q2" ///
3 "Destination Q3" 4 "Destination Q4") cols(2) size(small)) ///
xtitle("Years relative to first selected direct move") ///
yttitle("Mean log wage") ///
note("Balanced employed windows; descriptive selected-mover averages.") ///
xsize(9) ysize(5) name(fesim_cpv_movers, replace)
restore

```

6.11 A small estimation exercise

The simulator does not estimate AKM effects. Official Stata `areg` can fit a small connected example:

```

preserve
fesim, dgp(akmsimple) workers(500) firms(30) periods(6) ///
    seed(24680) truth(none) connectivity(largest) noreport clear
areg lnwage i.workerid i.time if employed, ///
    absorb(firmid) vce(cluster workerid)
restore

```

This dummy-variable fit is deliberately small. Estimation normalization is distinct from the latent-effect normalization. Connectedness alone does not remove limited-mobility bias in estimated variance components or establish the assumptions of a particular correction.

6.12 Three network experiments

```

preserve
fesim, dgp(akmsimple) network(blocks) workers(400) firms(40) ///
    periods(8) seed(13579) truth(full) ///
    parameters(block_count 4 block_log_bonus 2.1972245773362196) clear
tabulate worker_block_true firm_block_true if employed

fesim, dgp(akmsimple) network(bridges) workers(400) firms(40) ///
    periods(8) seed(246813) burnin(4) truth(full) ///
    parameters(block_count 4 p_eu .02 p_ee .60 p_ue .80) clear
matrix list r(bridges)

fesim, dgp(akmsimple) network(ladder) workers(500) firms(50) ///
    periods(8) seed(161803) truth(full) parameters(p_ee .30) clear
matrix list r(network)
matrix list r(leaveout)
restore

```

A Shared architecture and reproducibility

The ado layer resolves registry and options, validates support and dimensions, protects caller data/RNG, and dispatches to a Mata handler. The pipeline is: build population, initialize, discard burn-in, generate retained states, write the Stata panel, finalize flows, compute diagnostics, attach metadata.

- `fesim_registry.ado` defines parameter names, defaults, bounds, and provenance.
- `fesim_config.ado` adds cross-parameter restrictions and canonical serialization.
- `fesim_time.ado` validates dates and interval lengths.
- Numerical preset records live under `calibrations/`.

These implementation sources take precedence over historical roadmap descriptions of future options.

A.1 Independent random streams

`src/fesim_rng.mata` uses Stata’s `mt64s` with streams: 101 workers, 102 firms, 103 initial states, 104 mobility events, 105 destinations, 106 wage shocks, 107 observation error, 108 solver/calibration randomization, 109 network design. It stores full component states in Mata, restores the component before drawing, and restores the caller’s generator and state afterward. Reserved streams do not imply that every associated feature is public.

An explicit seed preserves the caller sequence. Without a seed, one caller draw selects a master seed in $[0, 2^{31} - 1]$. Reporting, truth selection, and deterministic output do not change economic paths. Invalid configurations fail before clearing or drawing; later errors restore the prior dataset and RNG.

A.2 Time and probability conversion

Let Δ be output-interval years. Simple AKM and pay-gap convert annual probabilities to interval probabilities:

$$P_{UE}(\Delta) = 1 - (1 - p_{UE})^\Delta, \quad (2)$$

$$A(\Delta) = 1 - (1 - p_{EU} - p_{EE})^\Delta, \quad (3)$$

$$P_{EU}(\Delta) = \frac{p_{EU}}{p_{EU} + p_{EE}} A(\Delta), \quad P_{EE}(\Delta) = \frac{p_{EE}}{p_{EU} + p_{EE}} A(\Delta). \quad (4)$$

When the denominator is zero, both competing probabilities are zero. At most one transition occurs per interval. This preserves the annual competing-risk interpretation but does not turn the discrete process into an exact continuous-time chain allowing repeated within-interval transitions.

Monthly AKM uses $d = 1/12$ and annual current-state hazards:

$$P_{EU}(d) = \frac{h_{EU}}{h_{EU} + h_{EE}} [1 - \exp\{-(h_{EU} + h_{EE})d\}], \quad (5)$$

$$P_{EE}(d) = \frac{h_{EE}}{h_{EU} + h_{EE}} [1 - \exp\{-(h_{EU} + h_{EE})d\}], \quad (6)$$

$$P_{UE}(d) = 1 - \exp(-h_{UE}d). \quad (7)$$

At most one event occurs per month; hazards update at the next monthly step. BM instead draws exact continuous waiting times.

A.3 Output and graph calculation

AKM handlers retain worker-length state and write period values directly into worker-major Stata rows without a duplicate full Mata panel. Complete-worker blocks finalize adjacent-snapshot flows. BM uses bounded history blocks (Appendix E.12). The finalizer computes moments, constructs unique observed matches, filters if requested, and recomputes moments and diagnostics on retained workers. Truth moments can exist when latent output columns are suppressed.

Weighting matters: primitive firm distributions weight each firm equally, whereas many wage/component moments weight employed worker-periods. Firm-size distributions use active firm-periods. Finite-sample reported moments therefore need not equal primitive parameters.

B Simple AKM implementation

B.1 Population and wages

`src/fesim_akm_simple.mata` draws independent standard normals a_i, z_j, v_j and sets

$$\alpha_i = \sigma_\alpha a_i, \quad \psi_j = \sigma_\psi z_j, \quad q_j = \frac{\exp(\sigma_q v_j)}{\sum_k \exp(\sigma_q v_k)}.$$

Weights use max-shifted logs floored at -700 to avoid numerical zero mass. Zero attraction scale gives exact uniform weights. Wage errors are independent $\epsilon_{it} = \sigma_\epsilon e_{it}$ in (1). One error draw per worker-period is consumed even if unemployed, preventing employment counts from shifting wage-stream positions. The wage trend starts at zero at the first retained observation.

B.2 Transition matrix and initial state

Write $a = P_{EU}(\Delta)$, $b_e = P_{EE}(\Delta)$, and $c = P_{UE}(\Delta)$. For random-network states $0, 1, \dots, J$, with 0 unemployed:

$$P_{00} = 1 - c, \quad P_{0j} = cq_j, \quad P_{j0} = a, \quad P_{jj} = 1 - a - b_e, \quad P_{jk} = b_e \frac{q_k}{1 - q_j} \quad (k \neq j).$$

The stationary initializer solves the actual interval transition rule (the random case uses a QR-based linear solve and a stationary-residual check). It does not assign employed workers simply in proportion to q_j when direct moves exclude the current firm. Positive job exits imply a stationary geometric backward age; if exits are zero, tenure is set to zero. Network-specific assignment restrictions enter their corresponding initial-state routines.

Random starts use one-half employment, attraction-weighted firms, and zero age. All-unemployed starts are draw-free. Burn-in repeats the interval rule. Simple AKM observes the initialized/burned-in state at the first retained row; subsequent rows advance one output step.

B.3 Sampling and spells

Each step draws one event uniform and one destination uniform per worker. UE samples entry weights; EE conditions on excluding the origin firm. EU preserves the completed internal spell count; UE and EE increment it and reset tenure. Stayers age by one interval. The output finalizer distinguishes changed spells from changed firm IDs.

C Monthly AKM and network implementation

C.1 Latent heterogeneity

Let a_i, v_i, z_j, ξ_j be independent standard normals before mixing. The worker mobility index is $m_i = \rho_{z\alpha} a_i + \sqrt{1 - \rho_{z\alpha}^2} v_i$, with wage effect $\sigma_\alpha a_i$. Standard-normal quintile cutoffs map m_i to $z_i \in \{-2, -1, 0, 1, 2\}/\sqrt{2}$. Population probabilities are equal; realized type counts are not forced to match.

Firm quality is the centered, sample-SD-standardized index $\rho_{q\psi} z_j + \sqrt{1 - \rho_{q\psi}^2} \xi_j$, with wage effect $\sigma_\psi z_j$. Correlation inputs govern latent normals, not exact realized correlations between discretized types and observed wages.

C.2 Hazards

For current firm quality Q_j , tenure τ in years, and unemployment duration d_u in years:

$$\log h_{EU} = \kappa_{EU} + \beta_{EU}^w z_i + \beta_{EU}^f Q_j + \beta_{EU}^d \log(1 + \tau), \quad (8)$$

$$\log h_{EE} = \kappa_{EE} + \beta_{EE}^w z_i + \beta_{EE}^f Q_j + \beta_{EE}^d \log(1 + \tau), \quad (9)$$

$$\log h_{UE} = \kappa_{UE} + \beta_{UE}^w z_i + \beta_{UE}^d \log(1 + d_u). \quad (10)$$

Slopes map to the `eu_*`, `ee_*`, and `ue_*` controls. Log-sum-exp stabilizes competing probabilities; small-exposure series avoid cancellation in $1 - e^{-x}$. State and durations update monthly.

C.3 Destination kernels and tables

For type z set $B_z = \theta_{\text{sort}} z + \theta_{\text{quality}}$. UE weights are $q_k \exp(B_z Q_k)$. EE weights from j to $k \neq j$ are

$$q_k \exp\{B_z Q_k + \theta_{\text{up}} \max(Q_k - Q_j, 0) + \theta_{\text{down}} \max(Q_j - Q_k, 0)\}. \quad (11)$$

A negative `theta_down` penalizes downward quality distances. The code sorts firms by quality and constructs stabilized cumulative tables by five types, rather than worker-by-firm matrices. Lower/upper ranges factor into origin terms and destination slopes $B_z - \theta_{\text{down}}$ and $B_z + \theta_{\text{up}}$. Binary searches and range sampling handle origin dependence and exclusion. Annual/quarterly/monthly snapshots advance 12/3/1 monthly steps, sum latent transitions, and convert year-valued durations to output units.

C.4 Calibration and stress designs

Germany fixes hazard/duration parameters at stylized values and fits only `theta_sort=2.2` after setting three dispersions. The recorded fit uses five calibration and five disjoint validation seeds,

100,000 workers, 10,000 firms, eight years, and five-year burn-in. Mean covariance is .0206703882 in calibration and .0200683908 in validation, against .0205 with absolute tolerance .0015. Interactive defaults use a smaller population.

Blocks multiply ordinary destination weights by a same-block factor. Bridge planning replays strict-block mobility on copied post-burn-in state and RNG records and chooses distinct eligible workers by independent network priorities. Economic simulation restarts from original records. An imposed bridge reuses its EE event's destination uniform and conditions the kernel on the target block. Later empirical hazards can respond to that employer: destination-only intervention does not imply invariance of all future events.

Ladder directions use tied wage-effect percentile ranks and a band. Direction shares are renormalized over positive ordinary destination mass, with ordinary weights within direction and the same destination uniform reused. Initial assignment, UE rules, wages, and the event rule are not independently redefined by the ladder option.

D Pay-gap implementation and decomposition

`src/fesim_paygap.mata` draws groups, group-scaled worker normals, common firm surplus, independent premium deviations, and separate attraction. The standardized worker wage primitive determines one of five quintile types. Conditional on group g and type z , destination weights are

$$\omega_{gj}(z) \propto q_j \exp\{(c_g + d_g z)s_j\},$$

where c_g and d_g denote `group_sort_g` and `worker_sort_g`. EE excludes the current firm. Group annual probabilities follow Appendix A.2. Random starts use one-half employment and group/type assignment weights before burn-in.

Let $\mathbb{E}_m$ and $\mathbb{E}_f$ average employed male and female observations:

$$D = (\mu_m - \mu_f) + (\mathbb{E}_m \alpha - \mathbb{E}_f \alpha) + F + T + E, \quad (12)$$

$$F = \mathbb{E}_m \psi_m - \mathbb{E}_f \psi_f, \quad T = \mathbb{E}_m(\gamma_m t) - \mathbb{E}_f(\gamma_f t), \quad E = \mathbb{E}_m \epsilon - \mathbb{E}_f \epsilon. \quad (13)$$

Here γ_g denotes the group-specific wage trend. Male-reference sorting and schedule terms are

$$S_m = \mathbb{E}_m \psi_m - \mathbb{E}_f \psi_m, \quad B_m = \mathbb{E}_f(\psi_m - \psi_f).$$

Female-reference terms are

$$S_f = \mathbb{E}_m \psi_f - \mathbb{E}_f \psi_f, \quad B_f = \mathbb{E}_m(\psi_m - \psi_f).$$

Both satisfy $F = S_g + B_g$. Symmetric terms are their averages. The nine returned rows comprise total, intercept, worker composition, firm total, sorting, premium schedule, time, residual, and adding-up error. Firm total is a subtotal; do not add it again to sorting and schedule.

CCK targets include male/female log-wage SDs .554/.513, worker SDs .420/.400, mean premiums .148/.099, premium SDs .247/.213, residual SDs .143/.125, and worker-premium correlations .167/.152. Targets for total gap, firm total, sorting, and schedule are .234, .049, .035, .015. Rounded sorting and schedule targets sum to .050, not .049; simulated decompositions obey their own exact identity. Female loading .12567 equals $.590 \times .213$, and its deviation SD is $\sqrt{.213^2 - .12567^2}$. The mapping targets selected moments under the package normalization, not the full CCK estimation procedure.

E Canonical Burdett–Mortensen implementation

E.1 Environment

Continuum workers and firms each have mass one. Firms have common flow productivity p and post a permanent wage w . Workers receive b while unemployed and discount at rate r . Unemployed/employed offers arrive at annual Poisson rates λ_u and λ_e , respectively; jobs are destroyed at rate δ . Offers sample firms uniformly. Firms maximize steady-state flow profit, without bargaining, counteroffers, endogenous wage revision, capacity constraints, or productivity heterogeneity. Unemployed workers accept wages weakly above reservation wage R ; employed workers accept only strictly higher wages.

E.2 Worker values and reservation

Let U be unemployment value, $W(w)$ job value, and F the offer CDF on $[R, \bar{w}]$. Bellman equations are

$$rU = b + \lambda_u \int_R^{\bar{w}} [W(x) - U] dF(x), \quad (14)$$

$$rW(w) = w + \delta[U - W(w)] + \lambda_e \int_w^{\bar{w}} [W(x) - W(w)] dF(x). \quad (15)$$

Reservation requires $W(R) = U$. Differentiating (15) gives

$$W'(w) = \frac{1}{r + \delta + \lambda_e[1 - F(w)]}.$$

Evaluating (15) at R , subtracting (14), and changing integration order yields

$$R = b + (\lambda_u - \lambda_e)S, \quad S = \int_R^{\bar{w}} \frac{1 - F(x)}{r + \delta + \lambda_e[1 - F(x)]} dx. \quad (16)$$

Thus $R = b$ when $\lambda_u = \lambda_e$. Imposing that equality with unequal rates generally violates the worker equations.

E.3 Stationary workers and firm labor supply

All support offers are acceptable in unemployment, so

$$u = \frac{\delta}{\delta + \lambda_u}, \quad e = 1 - u.$$

For employed-worker CDF G , inflows to jobs below w are $\lambda_u u F(w)$ and outflows are $e G(w) \{\delta + \lambda_e [1 - F(w)]\}$. Hence

$$G(w) = \frac{\delta F(w)}{\delta + \lambda_e [1 - F(w)]}. \quad (17)$$

Since $G(w) \leq F(w)$, workers concentrate at higher wages than offers. Employment per unit firm mass is

$$\ell(w) = e \frac{dG}{dF} = \frac{\delta \lambda_u (\delta + \lambda_e)}{(\delta + \lambda_u) \{\delta + \lambda_e [1 - F(w)]\}^2}. \quad (18)$$

Profit is $\pi(w) = (p - w)\ell(w)$. Continuum labor supply is not a simulated integer headcount.

E.4 Equal-profit equilibrium

Equating profit at w to profit at the lower support, where $F(R) = 0$, gives

$$F(w) = \frac{\delta + \lambda_e}{\lambda_e} \left(1 - \sqrt{\frac{p-w}{p-R}} \right), \quad (19)$$

$$\bar{w} = p - (p-R) \left(\frac{\delta}{\delta + \lambda_e} \right)^2, \quad (20)$$

$$w(q) = F^{-1}(q) = p - (p-R) \left(1 - \frac{\lambda_e}{\delta + \lambda_e} q \right)^2, \quad 0 \leq q \leq 1, \quad (21)$$

$$\bar{\pi} = \frac{\delta \lambda_u}{(\delta + \lambda_u)(\delta + \lambda_e)} (p-R). \quad (22)$$

Profit is constant on support. Wages below R recruit nobody; wages above $\bar{w}$ raise wage cost without adding recruitment or retention advantages over the top support wage.

E.5 Analytical reservation solution

Substitution of (19) into (16) gives $S = C(p-R)$:

$$C = \frac{2\lambda_e}{(\delta + \lambda_e)^2} \left[\frac{1}{2} - \frac{r}{\lambda_e} + \frac{r(r+\delta)}{\lambda_e^2} \log \left(1 + \frac{\lambda_e}{r+\delta} \right) \right], \quad (23)$$

$$R = \frac{b + (\lambda_u - \lambda_e)Cp}{1 + (\lambda_u - \lambda_e)C}. \quad (24)$$

This analytical expression is the primary solver. Denominator and support must satisfy interior and finite-precision checks. At public defaults, $C \simeq .92842983$, $R \simeq .59022409$, $\bar{w} \simeq .96654891$, $u = 1/6$, and $\bar{\pi} \simeq .09756569$. The continuum accepted EE hazard is

$$h_{EE}^\infty = \lambda_e \int_R^{\bar{w}} [1 - F(w)] dG(w) = \delta \left[\frac{\delta + \lambda_e}{\lambda_e} \log \left(1 + \frac{\lambda_e}{\delta} \right) - 1 \right], \quad (25)$$

approximately .15077363 at the defaults.

E.6 Worker values in full truth

For $q = F(w)$, the implemented values are

$$U = \frac{b + \lambda_u C(p-R)}{r}, \quad (26)$$

$$W(w) = U + \frac{2(p-R)\lambda_e}{(\delta + \lambda_e)^2} \left[q + \frac{r}{\lambda_e} \log \left(1 - \frac{\lambda_e q}{r + \delta + \lambda_e} \right) \right]. \quad (27)$$

These are *continuum* values evaluated at discretized wages, not newly solved Bellman values for the finite offer lottery. The simulation uses continuum support and strict wage comparisons; finite stationary employment is computed separately. See `src/fesim_bm_output.mata`.

E.7 Stability and residual checks

`src/fesim_bm.mata` solves afresh on each call. Public numerical controls are fixed: 1,001 wage-grid points, 4,000 even Simpson intervals, tolerance 10^{-9} . There are no public grid options or persistent caches.

For $x = \lambda_e/(r + \delta) < .1$, direct evaluation of the bracket in (23) subtracts large nearly equal terms. With $\rho = r/(r + \delta)$ the code uses

$$\frac{1 - \rho}{2} + \rho \sum_{n=1}^{30} \frac{(-1)^{n+1} x^n}{n + 2}.$$

Otherwise it uses $1/2 + \rho\{\log(1 + x)/x^2 - 1/x\}$. The logarithm helper uses a 40-term alternating series for $|x| < .1$. For $z = \lambda_e/\delta < .1$, the EE rate uses

$$h_{EE}^\infty = \delta \sum_{n=1}^{30} \frac{(-1)^{n+1} z^n}{n(n + 1)}.$$

Employment values use the stable logarithm helper. Exact CDF endpoints are explicitly preserved against division roundoff.

Independent Simpson quadrature over wages checks the reservation equation, scaling its residual by $\max(1, |b|, |p|, |R|)$. Equal-profit residuals are scaled by $\max(1, |\bar{\pi}|)$. Checks require finite support, strict $0 < R < \bar{w} < p$, positive employment/profit, ordered support, monotone CDFs in $[0, 1]$, and recomputed distribution/labor identities. Mathematically interior but numerically unresolved inputs can fail explicitly.

E.8 Finite firms and stationary masses

Default quantiles are $q_j = (j - \frac{1}{2})/J$ in (21). `random_firms=1` draws independent uniforms on stream 102 and sorts them. IDs increase with wage; original draw order breaks quantile ties. Each firm has contact probability $1/J$.

Let L_j be firm j 's share of *all workers*, $L_{<j}$ mass at strictly lower wages, and $H_j = \#\{k : w_k > w_j\}$. Exact finite flow balance is

$$L_j = \frac{\lambda_u u/J + \lambda_e L_{<j}/J}{\delta + \lambda_e H_j/J}. \quad (28)$$

The code processes equal-wage groups from low to high, using the same lower mass and higher-firm count within each group, then updating lower mass. Tied firms have equal masses and tied offers never induce EE moves. This linear pass avoids a dense transition matrix.

Masses sum to e and pass stationary-flow residual checks. Expected headcount is NL_j , conditional employed share L_j/e , and accepted finite EE hazard

$$h_{EE}^J = \lambda_e \frac{\sum_j L_j H_j/J}{e}.$$

Continuum-scale finite employment/profit are JL_j and $(p - w_j)JL_j$. They differ from NL_j and from continuum $\ell(w_j)$ and $\bar{\pi}$. Finite-scaled profits need not be exactly equal: wages discretize a continuum equilibrium rather than solve a finite strategic game.

For distinct midpoint wages, offer-CDF discrepancy is $1/(2J)$. Reported worker-CDF discrepancy compares finite cumulative conditional masses to (17) at grid points; it is not Monte Carlo error. More workers reduce sampling noise, whereas more firms refine discretization.

E.9 Initial states and burn-in

Stationary starts draw from $(u, L_1, \dots, L_J)$. Backward unemployment age is exponential at rate λ_u ; backward tenure conditional on firm j is exponential at rate $\delta + \lambda_e H_j/J$. Rejected offers do not reset tenure and do not enter this accepted-exit rate. One state and one age uniform per worker use stream 103.

Random starts put one-half probability on unemployment and one-half uniformly across firms, with zero ages. They require positive burn-in. All-unemployed starts have zero durations/spells and consume no draws. Burn-in runs the exact event engine without retaining the ledger, carries final states/ages forward, and resets retained counters. Exponential memorylessness permits a fresh wait at the burn-in boundary. See `src/fesim_bm_initial.mata`.

E.10 Exact event simulation

For each worker, `src/fesim_bm_events.mata` implements:

1. Set $h = \lambda_u$ if unemployed, or $\lambda_e + \delta$ if employed. Draw wait $-\log(U)/h$, mapping machine endpoint uniforms into the open interval so waits are strictly positive.
2. If arrival exceeds the horizon, age the state to the horizon and stop. Include an event exactly at the right boundary.
3. For an employed event, choose destruction with probability $\delta/(\lambda_e + \delta)$, otherwise an offer. Equality at the event-type threshold is an offer.
4. Contact a firm uniformly on any offer. Accept every support offer while unemployed, and only a strictly higher wage while employed. Current-firm, lower, and tied offers are recorded rejections.
5. Destruction resets unemployment age. Accepted UE/EE increments the worker-local spell and resets tenure. Continue from the new state.

Wait/type draws use stream 104; contacts use 105. The combined clock is equivalent to racing independent Poisson clocks. Workers do not compete for capacity, so worker-major generation is equivalent to calendar ordering across workers. Retaining the private ledger cannot alter paths or draws.

E.11 Observation aggregation

BM snapshots occur at $\Delta, 2\Delta, \dots, T\Delta$ after the initialization/burn-in boundary. The zero-time state is not returned. Intervals are right-closed: $((k-1)\Delta, k\Delta]$. An event at $k\Delta$ affects snapshot k and counts in interval k .

`src/fesim_bm_aggregate.mata` replays histories without draws, splits exposure at output boundaries, and counts unemployment/employed offers, rejections, destructions, accepted UE/EE, all events, and accepted state transitions. Employment plus unemployment exposure is Δ per row up to floating-point tolerance.

Endpoint flags are not event counts. Same-firm re-entry after destruction changes the spell but not firm ID. Re-entry elsewhere can yield observed `jobtojob=1` despite intervening unemployment. Forward-looking `to_unemp` places a separation flag on the origin row. For fixed seed, firms,

initialization, and total horizon, nested frequencies preserve final states, events, and exposure. Endpoint probabilities may differ, as Figure 6 illustrates.

E.12 Bounded blocks and resource guards

`src/fesim_bm_handler.mata` initializes the population, simulates complete worker histories in blocks, aggregates them, and writes directly into final Stata rows. Persistent event/destination buffers survive block boundaries and preserve the monolithic worker-major seeded path. The default block worker count is

$$\min \left\{ 10,000, \max \left(1, \left\lfloor \frac{100,000}{T} \right\rfloor \right), \max \left(1, \left\lfloor \frac{100,000}{1 + H \max(\lambda_u, \lambda_e + \delta)} \right\rfloor \right) \right\},$$

where T is periods and $H = T\Delta$ years. It targets roughly 100,000 rows/events per block; one long worker history can exceed the target. This is internal, not a public option.

Limits are 10 million workers, one million firms, 100,000 years per burn-in/retained horizon, and the common observation cap. Each burn-in and retained stage has a hard 10-million-event budget. Exceeding it causes rollback rather than truncation. No duplicate full Mata panel or retained full-population event ledger is stored. Initialization uses $O(N + J)$ storage and $O(N \log J + J)$ work; tie-aware exit rates are linear and offer acceptance is constant-time after validation.

E.13 Full truth and diagnostics

Basic truth stores log wage, accepted wage level, and productivity. No measurement error is added. Full truth adds:

Variables	Meaning
<code>reservation_wage_true,</code> <code>unemployment_value_true</code>	Continuum R, U , repeated on all rows.
<code>employment_value_true,</code> <code>offer_quantile_true</code>	Continuum $W(w_j)$ and accepted-firm offer quantile.
<code>expected_firm_mass_true,</code> <code>expected_firm_share_true</code>	Finite $L_j, L_j/e$.
<code>continuum_employment_true,</code> <code>finite_scaled_employment_true</code>	$\ell(w_j), JL_j$.
<code>continuum_profit_true,</code> <code>finite_scaled_profit_true</code>	$(p - w_j)\ell(w_j), (p - w_j)JL_j$.
<code>n_eu_true, n_ee_true, n_ue_true</code>	Accepted counts including the first interval.
<code>n_unemployment_offers_true,</code> <code>n_employed_offers_true,</code> <code>n_rejected_offers_true</code>	Primitive offer/rejection counts.
<code>n_events_true, ntransitions_true</code>	All events and accepted state transitions.
<code>employment_exposure_true,</code> <code>unemployment_exposure_true</code>	Exact years at risk in the interval.

Job/firm quantities are missing when unemployed. Counts and exposures describe intervals regardless

of endpoint employment. Internal spell counts survive unemployment; observed employment-spell IDs follow common missingness.

`r(solver)` has 22 rows: six primitives (using `discount` for r), surplus coefficient, reservation/upper wages, unemployment/employment, continuum/finite EE rates, equilibrium profit, unemployment value, raw/scaled reservation residuals, scaled equal-profit residual, finite stationary residual, offer/worker CDF errors, finite aggregate employment error. Its single column is `value`.

`r(bm_firms)` is 10-by-3 with `mean`, `min`, `max` across all firms, unweighted. Rows: firm ID, offer quantile, posted wage, expected mass, expected workers, expected share, continuum employment, finite-scaled employment, continuum profit, finite-scaled profit. It is a summary, not a J -row dataset.

`r(bm_flows)` has rows `unemployment_share`, `ue`, `eu`, `ee`, and four columns:

theory $u, \lambda_u, \delta, h_{EE}^J$ in the original finite economy.

event Unemployment-time share; UE count over unemployment years; destruction/EE counts over employed years.

observed Endpoint unemployment share and adjacent-endpoint transition probabilities.

observed_per_year Probabilities divided by Δ ; the unemployment-share row is missing. These annualizations are not hazards.

Absent at-risk exposure implies missing event rates. After `connectivity(largest)`, theory/-firm summaries still describe the original unconditioned economy. Realized moments, durations, event rates, and endpoint probabilities use retained complete-worker histories. Five temporary count/exposure columns enable this in every truth mode and are removed before return.

E.14 Validation scope

Tests cover equilibrium residuals, equal profit, Bellman identities, finite masses, ties, singleton firms, deterministic/random grids, replay, strict acceptance, burn-in convergence, truth/frequency/block invariance, rollback, and source-only installation. Statistical checks use fixed seeds and declared tolerances. The candidate inventory contains 68 Stata test files. Exact macOS Stata/MP 19 qualification receipts and archive hashes are recorded in the repository release-review report.

Limit tests approach equal offer rates, small employed offers, rare destruction, and small positive discounting. Public rates remain positive. A finite one-firm economy has no accepted EE events even though its parent continuum has positive EE mobility. Larger N cannot remove discretization error from fixed small J .

Historical benchmark source [38eb5bc](#) reports 100,000 workers over ten years in 17.943 seconds without truth and 21.526 seconds with full truth, with peak process RSS about 0.602 and 0.999 GB on that host. These environment-specific measurements are not guarantees. Public solve-only/saved-solution interfaces and heterogeneous BM remain outside the current command.

F CPV sequential bargaining implementation

F.1 Environment and the source protocol

Workers have fixed ability $\varepsilon_i > 0$; firm j has productivity $p_j > 0$. Output is $\varepsilon_i p_j$, unemployment flow utility is $\varepsilon_i b$, and utility is linear. The contact rates are λ_u while unemployed and λ_e while employed; jobs end at rate δ and workers discount at r . All firms have contact weight $a_j = 1/J$. A contact with one's employer is a null offer, not another competitor. Equal-productivity distinct firms can compete over wages, but never generate a direct employer move between themselves.

The protocol is CPV's original competition followed by bargaining over worker values. A less productive rival can trigger a raise even if its full value falls short of the worker's current promised value. Requiring that rival to beat the current promise would change the model. The share β is imposed, rather than estimated. At $\beta = 0$ competition operates as sequential auctions; at $\beta = 1$ the worker receives full match output and incumbent offers cannot produce further raises.

F.2 Finite-firm full values and unemployment

All following quantities are per efficiency unit. Let U be the value of unemployment, $B = rU$, and M_j the worker's value at firm j when the current wage equals its entire productivity p_j . Thus M_j is the full-productivity worker benchmark, not a separately computed sum of worker and firm values. Write $S_j = M_j - U$. An entry contract promises $U + \beta S_j$, giving

$$B = b + \lambda_u \beta \sum_j a_j S_j. \quad (29)$$

At full productivity an incumbent cannot improve the current contract. A higher-productivity rival k instead delivers $M_j + \beta(M_k - M_j)$. The worker's Bellman equation therefore implies

$$(r + \delta)S_j = p_j - B + \beta \lambda_e \sum_{p_k > p_j} a_k (S_k - S_j), \quad (30)$$

$$(r + \delta + \beta \lambda_e A_j^+) S_j = p_j - B + \beta \lambda_e T_j^+, \quad (31)$$

where $A_j^+ = \sum_{p_k > p_j} a_k$ and $T_j^+ = \sum_{p_k > p_j} a_k S_k$. Tails are strictly above the entire productivity tie group.

The implementation solves this finite system, rather than inserting continuum wages into a finite contact process. Starting with the most productive group, express each $S_j = A_j - BC_j$:

$$A_j = \frac{p_j + \beta \lambda_e \sum_{p_k > p_j} a_k A_k}{r + \delta + \beta \lambda_e A_j^+}, \quad C_j = \frac{1 + \beta \lambda_e \sum_{p_k > p_j} a_k C_k}{r + \delta + \beta \lambda_e A_j^+}, \quad (32)$$

$$B = \frac{b + \lambda_u \beta \sum_j a_j A_j}{1 + \lambda_u \beta \sum_j a_j C_j}. \quad (33)$$

Group suffix sums give linear work and storage after sorting. There is no dense J -by- J production solver. Solved surpluses must all be positive. The scaled residual of each Bellman equation, normalized by $\max(1, |p_j|, |B|)$, and the unemployment residual normalized by $\max(1, |B|)$ must be at most 10^{-9} . Independent small-economy tests solve the full dense linear system with a 10^{-12} comparison bound.

F.3 The contract state and each offer case

A contract stores employer j and reference q . The reference $q = 0$ denotes unemployment, with $S_0 = 0$. Define the promised surplus

$$C(q, j) = \beta S_j + (1 - \beta) S_q. \quad (34)$$

Upon a contact from firm k , the deterministic update is

$$(j', q') = \begin{cases} (j, q), & k = j, \\ (k, j), & p_k > p_j, \\ (j, k), & k \neq j, p_k \leq p_j, S_k > S_q, \beta < 1, \\ (j, q), & \text{otherwise.} \end{cases} \quad (35)$$

Only the second case is an EE transition. Only the third is a renegotiation. Tied firms share the same full surplus, so a different firm of equal productivity can bring the worker to full surplus. Reference updates at $\beta = 1$ are unnecessary when they change no value; the stored reference then remains the entry reference or last employer from a move. Neither null offers nor raises reset job tenure.

F.4 Recovering wages from promises

Let C'_k be the promised surplus after the contact update above. While the current contract is unchanged its wage is constant. Its Bellman identity yields

$$w(q, j) = B + (r + \delta)C(q, j) - \lambda_e \sum_k a_k \{C'_k - C(q, j)\}. \quad (36)$$

Higher rivals contribute $\beta S_k + (1 - \beta) S_j - C(q, j)$. Lower or equal rivals with $S_k > S_q$ contribute $(1 - \beta)(S_k - S_q)$, except the current employer, whose contribution is zero. Prefix sums of surpluses and group endpoints evaluate both sums in constant time; the current employer's atom is explicitly subtracted from the second sum.

Equation (36) gives $w(j, j) = p_j$ and an increasing wage in the reference surplus. Entry $q = 0$ is the minimum contract wage at a firm. Every entry wage must be positive; neither negative wages nor invalid surpluses are clipped. Higher worker value at a new employer need not mean a higher current wage, because the expected future gain term can increase. A frozen test verifies a value-improving move with a current wage cut, and another verifies a raise from a rival below the current promise.

As J grows, a uniform grid converges to the continuous offer model. For an interior reference x and employer $y > x$, the independent continuum check uses CPV's equation (3):

$$\phi(x, y) = y - (1 - \beta) \int_x^y \frac{r + \delta + \lambda_e \bar{F}(z)}{r + \delta + \beta \lambda_e \bar{F}(z)} dz. \quad (37)$$

For uniform productivity, the integral has an elementary logarithmic form (and a polynomial limit at $\beta = 0$). Tests compare it with 500- and 5,000-firm solutions away from the entry boundary. Finite self-contact and tied-firm corrections are retained in simulated wages.

F.5 Worker and firm primitives

For the simple preset, $\varepsilon_i = 1$. Otherwise,

$$z_i \sim N(0, 1), \quad \varepsilon_i = \exp(\sigma z_i - \sigma^2/2), \quad \sigma = \text{sd_worker}. \quad (38)$$

This is a population mean-one normalization, not sample restandardization. Every wage and value is multiplied by ε_i ; production is $\varepsilon_i p_j$. Nonpositive or nonfinite simulated ability or wages cause a transactional error. Because contact rates and comparisons are common and per efficiency unit, changing σ leaves the same seeded employment path intact. In population, ability is independent of employer productivity.

Midpoint firms use $v_j = (j - 1/2)/J$ and $p_j = p_{\min} + (p_{\max} - p_{\min})v_j$. Random firms use sorted independent uniform quantiles. IDs follow sorted productivity; exact ties remain ties. The economy does not add free entry, minimum wages, sector calibration, endogenous ability sorting, amenities, measurement error, or productivity shocks.

F.6 Exact finite stationary employment and raises

Unemployment is $u = \delta/(\delta + \lambda_u)$. If F is cumulative offer mass through a whole productivity group, stationary balance for workers at or below that group gives

$$G(F) = \Pr(p_{\text{employer}} \leq p \mid E) = \frac{\delta F}{\delta + \lambda_e(1 - F)}. \quad (39)$$

Divide each group increment in G equally among its firms and multiply by $1 - u$ to obtain unconditional masses m_j . The exact EE hazard per employment year is

$$h_{EE} = \lambda_e \sum_j \frac{m_j}{1 - u} A_j^+. \quad (40)$$

For an offered productivity group whose first sorted index is l , consider the stationary mass at any eligible incumbent with reference strictly below that offered productivity. Inflow is a UE into that incumbent or a move from a lower-productivity employer. Destruction or a sufficiently good rival removes the worker from this subset. Thus

$$H_l = \frac{u\lambda_u + \lambda_e \sum_{p_h < p_l} m_h}{J\delta + \lambda_e(J - l)}. \quad (41)$$

For each offered firm there are $J - l$ eligible incumbents, after excluding self. When $\beta < 1$, the exact raise hazard is

$$h_R = \frac{\lambda_e}{J(1 - u)} \sum_{\text{groups } g} |g|(J - l_g) H_{l_g}. \quad (42)$$

It is zero at $\beta = 1$. These quantities require linear work; a dense CTMC over all admissible (j, q) states is used only as an independent tiny-economy test oracle.

F.7 Stationary contracts and tenure without a dense state solver

A stationary employed worker's elapsed time since the last UE entry is $A \sim \text{Exp}(\delta)$: conditional on employment, entry cohorts survive at the common destruction rate. At time $-A$, sample a uniform

entry employer and set $q = 0$. Replay employed contacts at rate λ_e up to zero, conditioning on no intervening destruction. The final employer, reference and elapsed time since the last direct move jointly generate the stationary contract and job tenure. Sampling only an employer and assigning an independent entry wage would fail this joint-stationarity requirement.

Conditional unemployment, backward unemployment duration is $\text{Exp}(\lambda_u)$. All initialization draws, including historical offers, use the initial-state stream. The initial public spell counter starts at one for an employed worker; unobserved historical job numbers are not exposed. Random initialization instead uses half employment, uniform employers, entry contracts and zero ages. All-unemployed initialization uses zero ages and no initial-state draws.

Independent tests construct the finite contract-state generator and solve its stationary distribution. For each employer, uniformization of the within-employer subgenerator gives joint contract masses with job tenure exceeding two years. Two disjoint 100,000-worker fixed-seed samples must satisfy predeclared absolute bounds of .005 for state masses and .004 for the contract–tenure tail. Long burn-in and returned flow tests use separate seeds.

F.8 Event clock, observation, storage and failures

The total primitive event rate is λ_u while unemployed and $\delta + \lambda_e$ while employed. Waiting times are exponential. An employed event is destruction with probability $\delta/(\delta + \lambda_e)$; otherwise it is a uniform firm contact. Unemployed contacts enter a job, since every firm was validated viable. Self and rejected contacts remain primitive events. Destruction records an EU; entry records a UE; only an upward employer change records EE.

Snapshots are taken at $\Delta, 2\Delta, \dots, T\Delta$. An event exactly at the endpoint belongs to that interval and its resulting state is observed. A waiting time drawn beyond one endpoint is carried into the next interval. No redraw occurs just because output frequency changes. Exact E/U exposure accumulates along each segment in years. Observed flow flags follow the common endpoint rules, and public first-row `ntransitions` is missing even though full truth records that first interval's latent counts. A same-firm return creates a new spell and latent EU/UE without an observed employer-to-employer flag.

Uniform draws are buffered in blocks of 4,096 per component. Buffers survive worker and processing-block boundaries. Ability, firm primitives, initial states, event clocks and destinations use the established separate RNG streams. Fixed-seed none/basic/full output, ability changes, block sizes and nested frequencies are checked for path invariance.

The public writer uses at most $\min\{10,000, \max(1, \lfloor 100,000/T \rfloor)\}$ workers per block; a single worker's full requested history is the unavoidable minimum. It stores no population-wide event ledger. Bounds are ten million workers, one million firms, 2,147,483,647 requested rows, and 100,000 years for each horizon. Initialization permits at most ten million historical employed offers; burn-in and retained simulation each have a ten-million primitive-event budget. Failure restores caller data and RNG, including errors after random firm construction. Nonfinite lifetime values or ability-scaled productivity/value bounds are rejected under every truth mode.

F.9 Truth and result interpretation

Basic truth reports five structural objects:

Variable	Meaning
<code>lnwage_true</code>	Log contract wage
<code>contract_wage_true</code>	Contract wage level
<code>worker_ability_true</code>	Fixed worker efficiency units
<code>firm_productivity_true</code>	Firm productivity per efficiency unit
<code>match_productivity_true</code>	Worker ability times firm productivity

Ability is defined even while unemployed; job attributes are missing then. Full truth adds the reference firm, reference surplus and value, unemployment value, employment value, full-productivity match benchmark, and exact interval EU/EE/UE, offers, rejections, events, transitions, renegotiations and exposure. Reference firm zero means unemployment; reference value is then the worker's U . All reported values scale with worker ability.

`r(solver)` has 17 named rows: primitives and realized firm support, unemployment value and shares, finite EE and raise hazards, Bellman residual, minimum surplus and entry wage, and firm count. `r(cpv_firms)` summarizes eight firm objects by mean/minimum/maximum: ID, quantile, productivity, full surplus, entry wage, expected mass, expected workers and conditional employment share. No large public Stata matrix of all firms is required.

`r(cpv_flows)` has rows for unemployment share, UE, EU, EE and renegotiation, and columns for theory, event/exposure, observed endpoint probability and the descriptive endpoint probability per year. Renegotiation has no endpoint-transition analogue, so its last two columns are missing. With `connectivity(largest)`, theoretical objects and generated-event counts describe the original economy; realized rates, moments and durations are recomputed for returned workers. Six temporary count/exposure variables are removed before return under every truth setting. Structural truth is not labeled as AKM firm truth.

F.10 Equation audit and qualification boundary

The source audit uses the November 2005 author revision linked from Jean-Marc Robin's publications page, SHA256 prefix `f7e51c74dbd34914`. Rendered PDF pages 9–11 (printed 7–9) establish the bargaining cases, equation (3), beta limits and wage cuts. Its equation (4) visibly prints p_{inf} as the first term although it equates the expression with equation (3), which uses employer productivity p . This apparent typo is not copied: entry wages follow Equation (36). The exact publisher PDF has not been verified; version-specific source claims retain that qualification. Full source identity, URL and evidence are in `docs/cpv-derivation.md`.

The finite solver, continuous-time simulator and examples implement the canonical mechanisms with declared stylized primitives. They do not replicate empirical estimates or calibrate sector and skill groups. Qualification is for the tested macOS Stata/MP 19 environment. A release tag, publication and broader platforms remain separate decisions.

G Source map and rebuilding the manual

Source	Responsibility
<code>fesim.ado</code> , <code>fesim_config.ado</code> , <code>fesim_registry.ado</code>	Syntax, configuration, registry, metadata.
<code>src/fesim_akm_simple.mata</code> , <code>src/fesim_akm_handler.mata</code>	Simple-AKM primitives, transitions, wages, streaming.
<code>src/fesim_empirical.mata</code> , <code>src/fesim_hazards.mata</code> , <code>src/fesim_destinations.mata</code> , <code>src/fesim_emp_handler.mata</code>	Types, monthly hazards, destination tables, streaming.
<code>src/fesim_paygap.mata</code> , <code>src/fesim_paygap_handler.mata</code>	Group primitives, schedules, mobility, decomposition.
<code>src/fesim_network_design.mata</code> , <code>src/fesim_network.mata</code>	Stress designs and observed graph/audit.
<code>src/fesim_bm.mata</code>	Continuum solution and finite firms.
<code>src/fesim_bm_initial.mata</code> , <code>src/fesim_bm_events.mata</code>	Initial states, burn-in, exact events.
<code>src/fesim_bm_aggregate.mata</code> , <code>src/fesim_bm_output.mata</code> , <code>src/fesim_bm_handler.mata</code>	BM aggregation, truth, blocks, diagnostics.
<code>src/fesim_cpv.mata</code> , <code>src/fesim_cpv_simulate.mata</code> , <code>_fesim_cpv.ado</code>	CPV values, contracts, exact paths and public transaction.
<code>src/fesim_rng.mata</code> , <code>src/fesim_flows.mata</code> , <code>_fesim_finalize.ado</code>	RNG, common flows, result finalization.

From the repository root, regenerate figures in Stata:

```
do docs/manual_figures.do
```

The driver runs ten examples and exports vector PDFs to `docs/manual_figures/`. Each example also runs independently. Fixed-seed public simulations supply plotted data; analytical comparisons use the formulas in this manual. Raw simulated data are not needed to compile the document. From a shell at the repository root:

```
bash docs/build_manual.sh
```

This requires `latexmk` and `pdflatex`. Auxiliary files go to `build/manual/`; the finished PDF is `docs/fesim_manual.pdf`. TeX is a documentation dependency, not a simulation dependency.

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